Introduction

In 2017, US News and World Report published an article "Healthcare will Bankrupt the Nation" [1]. Others were saying that healthcare would bankrupt every country. Healthcare was running at 17%–18% of US GDP and growing, although other countries seemed to deliver better outcomes with 9%–11%, although growing. And still, many United States residents were without healthcare and could be bankrupted by a serious disease. The evidence pointed to a need for improvement in processes and the delivery of more effective care. Not that other countries were immune. Aging demographics were driving rapidly increasing costs in every country's healthcare system.

Then, in 2019, came an announcement came that shook the industry. Amazon, JPMorgan Chase, and Berkshire Hathaway announced that they were entering healthcare. It seemed like a good idea at the time. They named the new company Haven Healthcare and appointed as CEO Dr. Atul Gawande. Gawande is a famous surgeon and gifted writer and explainer—his book "Being Mortal" showed heart and expertise in both living and healing.

The incumbent healthcare industry was especially shaken by the announcement of Haven. This new trio brought a tremendous history of leadership and winning. Amazon had already transformed the retail industry, had unmatched operational excellence, and was also a tech super-force as the dominant provider of the leading cloud computing platform Amazon Web Services that was transforming the IT industry with a cost-effective rent-versus-buy model. JPMorgan Chase was the largest US bank, one of few that had sailed through the Global Financial Crisis of 2007–08—they certainly seemed to be savvy and knew how to handle risk. And Berkshire Hathaway was the company of the most famed stock-picker for generations, Warren Buffet. A grandfatherly figure, he generously advised people even when it did not benefit his company—he told investors to just buy the S&P 500—that turned out to be some of the best advice for the era. But he also said, "*The ballooning costs of healthcare act as a hungry tapeworm on the American economy.*"

Haven Healthcare's idea [2] was that they would first serve the 1.1 million employees of the three companies, a captive audience and great "starter set." But JPMorgan Chase CEO Jamie Dimon was saying it *could eventually benefit all Americans*, stoking fear in healthcare incumbents.

But then, in Jan 2021, Haven Health abruptly announced closure of operations, only saying that the companies "*will leverage the insights and continue to collaborate informally to design programs tailored to address the specific needs of their own employee populations*" [3–5]. Haven has never fully explained its reasons, but this kind of "window dressing" is how companies usually deliver bad news. Amazon did go on to acquire One Medical, a membership-based primary care provider.

A somewhat parallel path was playing out at Walmart, Amazon's nemesis in the retail business and a well-run company. Walmart saw the attraction of this large potential market. Walmart is already at a well-run company. Walmart saw the attraction of this large potential market. Walmart is already at "the edge," with over 10,000 stores in the US, and they touch millions of people as they go about their lives. For many rural residents in the US, Walmart is the only "institution" that they regularly contact. Walmart opened 51 health centers, mainly in five southern US states, locating them near their megastores, and soon announced plans for expansion. They also provided virtual care through telehealth. While these offerings seemed to have promising uptake during the pandemic, in April 2024 Walmart suddenly announced closure of all 51 centers and an exit from virtual care. In announcing [6] their closing of 51 centers across five states, Walmart simply said "Through our experience managing Walmart Health centers and Walmart Health Virtual Care, we determined there is not a sustainable business model for us to continue." Later, in July 2024, CenterWell, insurer Humana's healthcare

services business, announced plans to lease the space next to Walmart stores that were formerly Walmart Health locations and open 23 senior-focused primary care centers [7].

It was not just newcomers to healthcare. United Healthcare (UHC), the US's largest insurer-provider with over 90,000 employed or associated doctors in its Optum subsidiary, also tried its hand with its Optum Virtual Care in 2021, and then in April 2024 announced its closure.

How could these very successful businesses make such radical reversals in the space of a few years? For Haven's failure, various reasons were given including their inability to get a scale that gave market purchasing power (Amazon's specialty, although this is surprising with 1.1 million lives being managed), COVID-19, the complexities of the business, and the difficulties of a fixed price model (such as fee-for-membership, discussed in Chapter 1) and the risks and startup costs that it entails. But the interview with Gawande in Fast Company [3] throws some light. Gawande noted that you had to pull four streams together to create change: access to patients, access to clinicians, access to setting incentives, and access to data. This starts to show the complexity of the task and, as he said, "you're either small and have control of all four or to get to scale you have to learn to partner." But it also either small and have control of all four or to get to scale you have to learn to partner." But it also confirms the fact that optimizing, and indeed transforming, healthcare requires simultaneously juggling a lot of both business and technical issues: primarily business model design but aided by tech. These reversals show that healthcare's complexity is poorly understood. To put it simply, healthcare is hard. Unwrapping some of that complexity and explaining some of the best current and potential approaches will be a theme of this book.

Note that both Walmart and Haven cited *business model* challenges, which may be understandable given the general acknowledgment of increasing costs and complexity of care. But there also pervasive problems and underlying needs in the *business process*, that is, the way healthcare is carried out. We will cite just two of them:

1. The complexity resting on the clinician's shoulders is enormous and getting worse every day. The cognitive overload of having to digest the history of the patient, their medical records and family history, their labs and imaging studies, and understanding the patient and their needs and how much they can care for themselves, is overwhelming, especially in a 10-to-20-minute encounter. That is to say nothing of keeping up with medical literature and new findings or evidence for the best treatments. The net result of this is evidenced in systematic studies such as the RAND Corporation's [8] which shows that, across a variety of diseases, compliance of clinicians with treatment guidelines ranges through 40%–60% and averages about 55%. A similar set of studies from developing countries on the know-do gap [9] show that clinicians usually "know" what to do about 70% of the time but are only able to "do" that about 40% of the time. Of course, there are times when departure from guidelines and usual treatments are desirable, and this is the role of the clinician to decide, but we are giving today's clinicians a herculean task. And then we ask them to document everything in an electronic health record (EHR) system and help the patient get insurance approval for their needed treatments. These workloads are generally acknowledged to be too burdensome to be compatible with optimal care—something has to change.

2. The second major observation is the 40/30/20/10 rule which we explore more in Chapter 2. Basically, it says that only about 10% of a patient's outcomes is controlled by what happened in the doctor's office anyway! Lifestyle, environmental, and social factors are the dominant determining factors of patient outcomes. The implication is that other longer-term and broader transformations, and new ways of reaching and changing patients' behaviors while at home and in the community, are needed to make a serious dent in outcomes and costs.

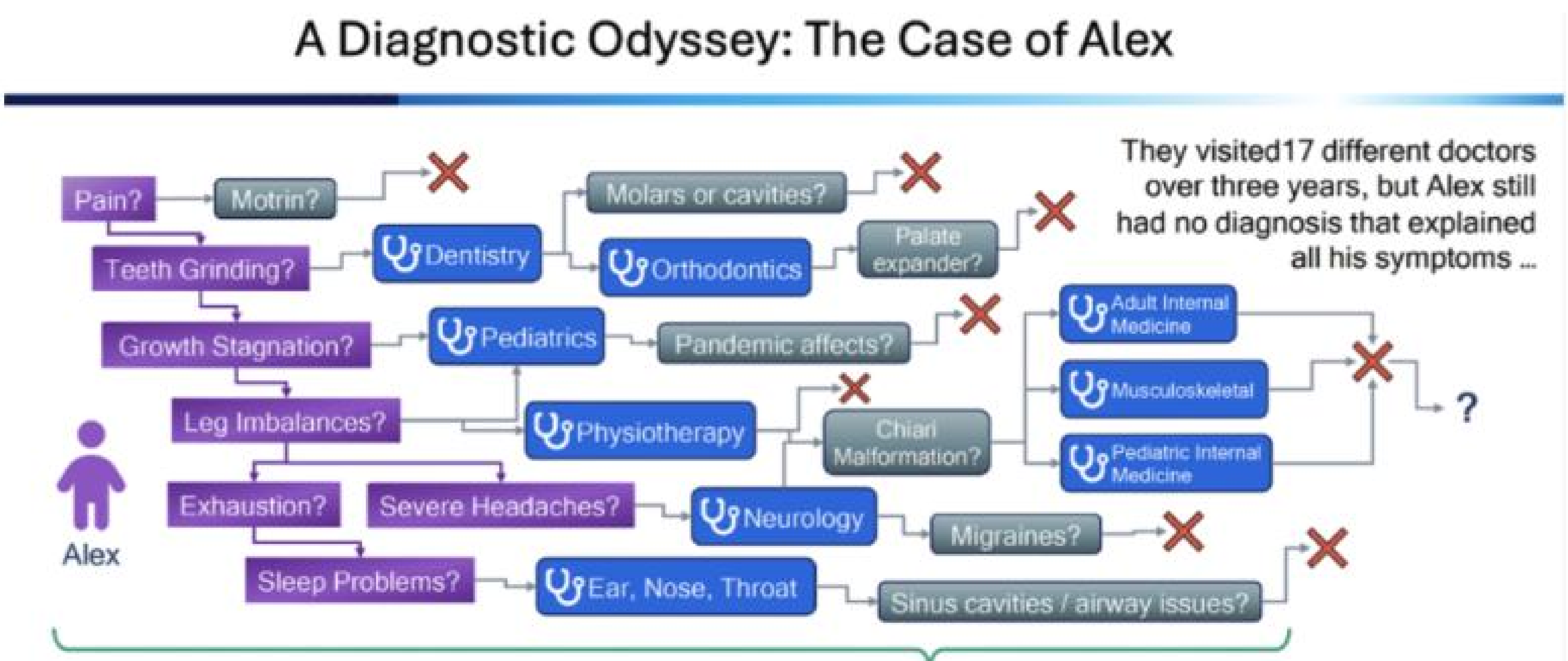
There are hints that the failure of the above companies' plays may have demonstrated a failure to embrace the importance of taking this much broader, holistic, and more comprehensive view of how to transform. This includes longitudinal and longer-term views of patient outcomes. As we will see in Part 3, chronic diseases, as the largest precursors of major disease, need to be the target of sustained and nonepisodic approaches. The same is true of hospitalization which incurs the largest part of overall costs.

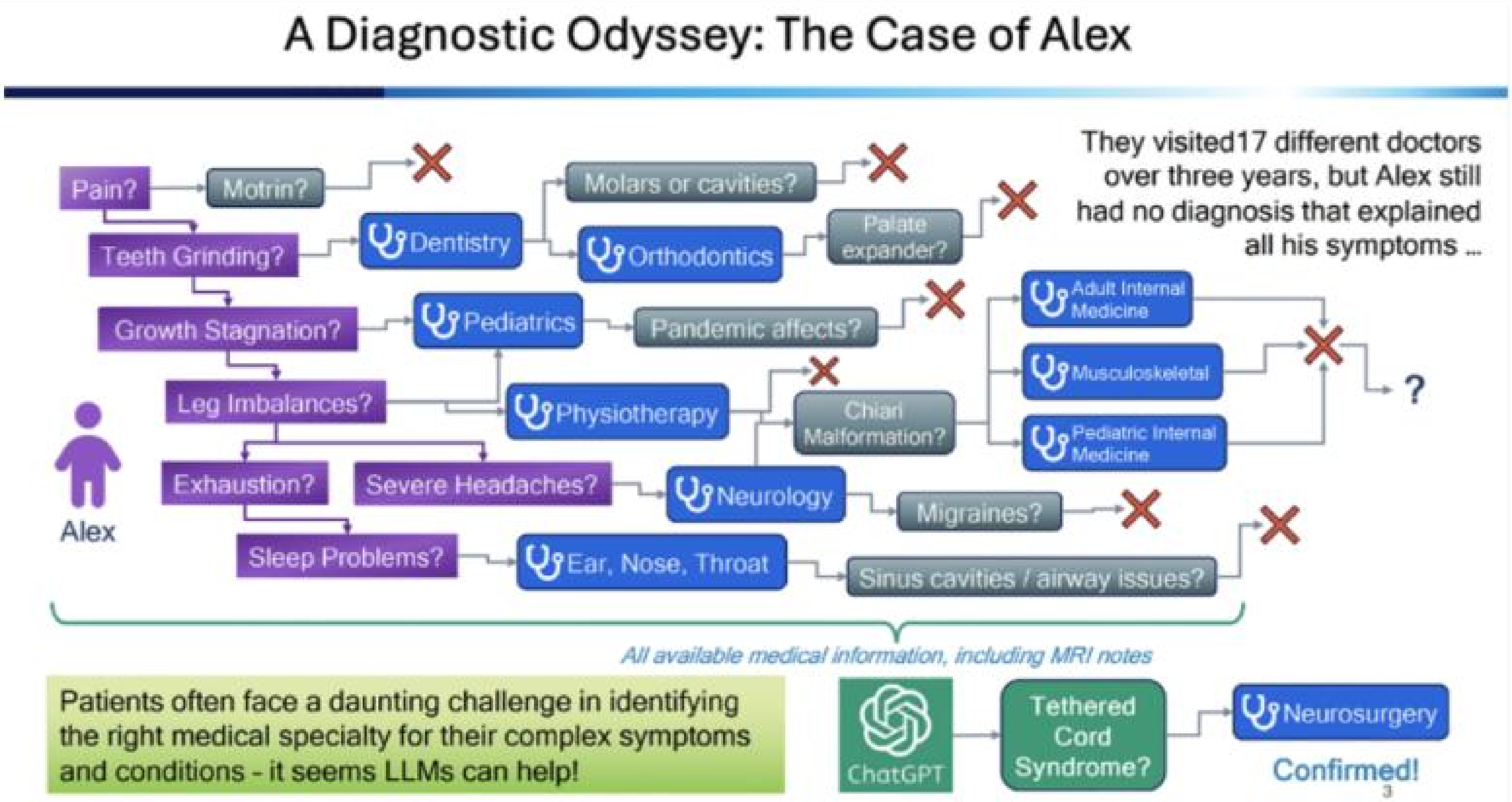
Telehealth seemed like a good substitutional lever, what happened? It was undoubtedly useful during COVID-19 where it experienced a surge because patients either could not or should not have gone into the clinic or hospital. But as things got back to normal, it was also realized that telehealth was not a cost saver (this was already well established, see data cited in Chapter 3): the same amount of time had to be spent by the clinician whether he or she was face-to-face or on the screen, maybe even more with the fiddling around and scheduling of the virtual call versus people already in the waiting room.

What is clear is that pure tech plays such as telehealth would not fundamentally change the game.

And about this time, something else was happening on the tech scene. At the end of 2022, ChatGPT was announced and had the fastest uptake of any technology yet. Peter Lee and his team at Microsoft Research had had early access to ChatGPT 4 and quickly released an intriguing book [10] that outlined the potential of generative AI in healthcare. But they also took pains to illustrate its weaknesses and how it could sometimes "hallucinate" and give incorrect or made-up material in answers. But there might be something there.

Then, in September 2023, the website of the TODAY show reported [11] a remarkable health story. A 4-year-old boy, Alex, who had previously been well and normal, started chewing on things and having meltdowns. The boy's dentist ruled everything out, but then the boy stopped growing and started dragging his left foot. No matter how many doctors the family saw, the specialist would typically look for something within their area of expertise and then rule it out. The family visited 17 different doctors over 3 years, but there was still no diagnosis that fitted with his symptoms. Then, the mother went into ChatGPT and entered everything from symptoms to MRI notes. Eventually ChatGPT suggested *tethered cord syndrome*, where tissue in the spinal cord attaches itself to the vertebrae suggested *tethered cord syndrome*, where tissue in the spinal cord attaches itself to the vertebrae limiting movement. The mother went to a new neurosurgeon and told her that she suspected tethered cord syndrome. The neurosurgeon looked at the images and immediately said "Here's *occulta spina bifida* and here's where the spine is tethered!" The family's diagnostic Odyssey is illustrated in the figure below. After receiving the diagnosis, Alex underwent surgery and was soon recovering well. While this is just one case, it illustrates the difficulty and complexity of diagnosis, especially for less common diseases, and that some way of digesting all the information that is out there might have a role to play.





A Diagnostic Odyssey: Case involving Tethered Cord syndrome [11]. Figure by Hua Xu, as presented at AI in Health Conference, Rice University, Sep 10–11, 2024 [12].

Technology had already made some good strides in healthcare such as in digitalizing health records (EHR) and automating some basic functions (e.g., linking to billing systems). However, it was also commonly noted that "feeding the EHR beast" was causing clinicians to spend too much time facing the screen and diverting time and attention away from the patient. And some of the initial attempts at applying AI to healthcare had not worked out too well. Radiology was an exception, where AI-based image processing was performing remarkably well, as we shall see in Chapter 12. But now, with generative AI, a completely new class of technologies was on the scene that would require a reckoning.

This book is about healthcare, what is good and what is in need of improvement, its methods of transformation, and about some relatively successful transformative projects around the world. These innovations may involve removing inefficiencies, changing the location of care, and using technology for greater efficiencies and quality. In a few cases, they are fundamentally changing the business model. While some are succeeding at bending the curve of increasing costs, most importantly they are improving patient outcomes. The net is that people will be able to lead healthier lives, get sick less often, and recover better when they do.

often, and recover better when they do.

It is also about AI. AI has been around for more than 50 years in various forms. It has been through fits and restarts, sometimes referred to as AI Winters and AI Springs. But in the last 10 years, we have seen a steady stream of more impressive and fundamental advances in AI. These have caused step changes in, to give a few examples, how well AI could recognize images, understand speech, or translate languages. Then, in late 2022, breakthroughs in generative AI burst upon the scene, with names like ChatGPT, Bard, Claude, LLaMA, Dall-E, etc.

What was amazing about these new tools was that they could hold a passable human-like conversation and summon up vast expertise (with occasional outright errors, also like humans) and were

easy to use through a conversational *bot* interface. They could also compose images from a brief prompt by the user describing what they were looking for. We will explain later in Part 2 how various kinds of AI systems work.

At the same time, medical science has been regularly exhibiting miracles with new drugs, vaccines developed in record times, etc. Meanwhile, the healthcare system has been groaning under ever-increasing costs and waiting lines, staffing shortages, higher costs to the end user, and unequal access both within countries and across countries. Aging is a major contributor to the increasing loads on the systems, since a large portion of the healthcare spend is on aged patients. We will show, by close examination, that a lot of the unequal quality, and increasing costs result from an increasingly complex job for clinicians. While clinicians generally do an excellent job, all health workers are confronted with massive information overload, whether this results from understanding patients and their histories, what treatments are out there and how to best match them up to the patient, and then how to deal with administrative burdens and record keeping, helping the patients with reimbursements, etc.

Could healthcare and tech be a marriage made in heaven? The marriage would be between what is basically an information-intensive enterprise and the availability of a new class of machines. These

Could healthcare and tech be a marriage made in heaven? The marriage would be between what is basically an information-intensive enterprise and the availability of a new class of machines. These new machines would help manage that information, get it in the right place at the right time, filter and summarize it to prevent information overloads, and then in some cases, automate or assist with timesaving and productivity improvements in the actions of the clinical care provider. And it could also be a way for patients to be much more aware of their own health-impacting behaviors, to listen to their bodies, and to self-manage some illnesses when they occur.

Indeed, this information revolution in healthcare had already been ongoing at a steady pace for over 30 years. EHR systems increasingly provide shortcuts and added functions such as helping with care advisories, summarizing recordings of clinical encounters, and then following up with patients when they are at home. These systems are also helping with some aspects of billing—collecting revenue is essential for healthcare systems to keep going and stay in business.

But AI was starting to offer fundamentally new functions. For example, in medical imaging, AI-based systems were starting to read images quite accurately and help the radiologist avoid missing certain findings and then summarize the findings of an imaging study. In some aspects of image interpretation, AI was comparable in skill to radiologists, but the combination of human and machine was still far superior because doctors are experienced with progression of disease, combining diverse sources of information about patients' histories, etc.

As we will see in the introductory chapters, other changes were afoot. There has been a beneficial shift in outcomes by involving patients more in their prevention and recovery. The strong evidence that the largest part of medical outcomes is determined by how patients lead their lives, manage their disease once diagnosed, etc., is inspiring new approaches. Hence, new methods such a remote (home) monitoring, coaching, and appropriate telecare are progressing. These trends have been a positive side-effect of the COVID-19 pandemic, where remote care methods became more accepted, even though subsequent experience showed that they need to be carefully and selectively applied. Accompanying this has also been breakthroughs in hospital-at-home, where patients' hospital admissions could be avoided, and patients could be discharged earlier.

The scope and extent of information and the amount of data itself were rapidly expanding, and what better to manage it than the increasingly sophisticated methods of data management including AI.

In this book, we will trace the progress of movements of healthcare model transformation and, in parallel, the progress in AI. We will see how these two streams are converging into a holistic mix that offers the potential of making healthcare better for all of us: the patient, the provider; the old and the young, rich and developing countries, the well, and the sick. All this with an important and essential side-benefit: the management of costs. As the world ages, healthcare costs have been growing without bound. It is not only hoped but necessary that technology and AI will eventually increase both efficacy and efficiency and thereby keep cost growth in check.

All this does not come without risks. The technology and the AI offer great potential but must be introduced and deployed in a way that does this with as little risk and downsides as possible. When GenAI is trialed in a tech support or business contact center, we can be quite exploratory, as the risks of an incorrect recommendation usually have limited impact. But using GenAI in medicine is something altogether different. We must be very careful when we introduce GenAI bots directly to patients, as there could be serious safety implications. An enterprise as complex as healthcare always comes with some errors and oversights, sometimes human errors, and sometimes errors in management processes and systems. While baseline medical error rates are already higher than we would like, we must be even more careful whenever we leave decisions to machines. By now, we have all experienced the "cleverness" of GenAI but also its hallucination errors. Both self-care and medicine are fundamentally human endeavors, all about caution, trust, empathy, and hope. As I have journeyed through healthcare settings across multiple countries, the most common view I have heard expressed is excitement and hope: that we will find ways for humans and machines to work together in creating positive transformations toward better health.

References

[1] Health Care Will Bankrupt the Nation. https://www.usnews.com/opinion/articles/2017-07-14/health-care-will-bankrupt-the-nation-we-need-reform-now.

[2] Haven Healthcare. https://en.wikipedia.org/wiki/Haven_Healthcare.

[3] Atul Gawande: To fix our broken healthcare system, start with primary care. https://www.fastcompany.com/90590937/atul-gawande-interview-haven-healthcare.

[4] What Does Haven Healthcare's Failure Say About Healthcare Innovation? https://www.linkedin.com/pulse/what-does-haven-healthcares-failure-say-healthcare-james-jordan/.

[5] What Does Haven Healthcare's Failure Say About Healthcare Innovation? https://www.linkedin.com/pulse/what-does-haven-healthcares-failure-say-healthcare-james-jordan/.

[6] Walmart Health Is Closing. https://corporate.walmart.com/news/2024/04/30/walmart-health-is-closing.

[7] J. Lagasse, Humana's CenterWell opening 23 primary care clinics in Walmart collaboration, 2025. https://www.healthcarefinancenews.com/news/humanas-centerwell-opening-23-primary-care-clinics-walmart-collaboration.

[8] E.A. McGlynn, et al., The quality of health care delivered to adults in the United States, New Engl. J. Med.

References

[1] Health Care Will Bankrupt the Nation. https://www.usnews.com/opinion/articles/2017-07-14/health-care-will-bankrupt-the-nation-we-need-reform-now.

[2] Haven Healthcare. https://en.wikipedia.org/wiki/Haven_Healthcare.

[3] Atul Gawande: To fix our broken healthcare system, start with primary care. https://www.fastcompany.com/90590937/atul-gawande-interview-haven-healthcare.

[4] What Does Haven Healthcare's Failure Say About Healthcare Innovation? https://www.linkedin.com/pulse/what-does-haven-healthcares-failure-say-healthcare-james-jordan/.

[5] What Does Haven Healthcare's Failure Say About Healthcare Innovation? https://www.linkedin.com/pulse/what-does-haven-healthcares-failure-say-healthcare-james-jordan/.

[6] Walmart Health Is Closing. https://corporate.walmart.com/news/2024/04/30/walmart-health-is-closing.

[7] J. Lagasse, Humana's CenterWell opening 23 primary care clinics in Walmart collaboration, 2025. https://www.healthcarefinancenews.com/news/humanas-centerwell-opening-23-primary-care-clinics-walmart-collaboration.

[8] E.A. McGlynn, et al., The quality of health care delivered to adults in the United States, New Engl. J. Med. 348 (26) (2003) 2635–2645, https://doi.org/10.1056/NEJMsa022615.

[9] Poor Countries It is Easier than Ever to See a Medic. https://www.economist.com/international/2017/08/24/in-poor-countries-it-is-easier-than-ever-to-see-a-medic.

[10] P. Lee, C. Goldberg, I. Kohane, The AI revolution in medicine: GPT-4 and beyond. Pearson, 2023.

[11] M. Holohan, A boy saw 17 doctors over 3 years for chronic pain, ChatGPT Found the Diagnosis (2025). https://www.today.com/health/mom-chatgpt-diagnosis-pain-rcna101843.

[12] H. Xu, Large language models for biomedical applications, AI in Health Conference, Rice University, Houston, Sep. 10, 2024. Accessed: Jan. 17, 2025 [Online]. Available: https://www.youtube.com/watch?v=NHAgVKQbtK8&list=PLcsG4X8Zn_UDLZgXYunmn-Z3HKkyMXynH.

CHAPTER

The healthcare industry

1

This book is about transforming healthcare using AI. It is not about an AI hammer looking for a nail in healthcare to batter down. So, we need to first define and understand the most relevant aspects of health and healthcare, locate pain points and inefficient parts of healthcare systems, and identify where outcomes could be improved. In looking for instructive examples, we are fortunate that some countries, and some private entities, have independently put together healthcare provider models that have been tested in various settings and have delivered improved outcomes and costs. We will identify some fundamental challenges within healthcare, and this will start to show how emerging technologies and AI may have a significant role to play in delivering improvements.

Several industries have been through fundamental transformations in the last few generations, and it will be instructive to see what kind of general transformational techniques have been successful. There are changes in the business processes, for example changing the location of the delivery of a service that may lead to an improvement. In healthcare, this might translate into moving care from a hospital or clinic to the home. Other industries tell the story of how inefficiencies and bottlenecks can be relieved or inefficient use of resources can be remedied. There are instances where quality defects are the problem and there are some simple and powerful lessons in how quality can be improved. And there are different financial models where innovative methods of charging or paying for services make the difference. An example in healthcare may be incentives for prevention. And there are cases where having the right information at the right moment is a valuable lever.

Technology and information can play a fundamental role when it enables process improvements that lead to better outcomes. This is sometimes done by implementing structural changes, that is, doing things a different way. But information can also improve systems more *parametrically*. This could be systematized through a methodology of *continuous improvement*, which could involve successively reducing error rates or waste, or improving resource allocation, for example, smoothing out demand, reducing idle resources or excessive waiting times, allocating scarce resources to the neediest cases, reducing idle resources or excessive waiting times, allocating scarce resources to the neediest cases, etc. As we shall see, for the case of healthcare, one key opportunity is in *decision support* at the point of care. The idea is to help by providing the right information just-in-time in the form of reminders and care suggestions, together with some automation. However, the news is also full of bitter complaints about electronic health record (EHR) systems that require extra work by clinicians, avert their eyes away from patients to screens, and require "pajama time" to catch up with all the data entry at home after a long workday. EHRs are claimed to be prime contributors to clinician burnout [1,2].

Unwise uses of technology and automation can also backfire on us, create extra work or distractions for precious staff, increase costs, or disrupt processes by concentrating too much on solving a problem which is not a limiting constraint within the system. The current technologies of greatest interest and

Healthcare Transformation using Artificial Intelligence. https://doi.org/10.1016/B978-0-443-28969-9.00001-7

applicability are mobile, data, analytics and, of course, AI. The potential of tech appears great, but how can it be carefully, tastefully, and safely applied when and where it is most needed?

In the transformation of any enterprise, we begin with the definition of the problem and the scope of the system being addressed. We then conduct analysis and diagnosis, consider instructive success stories, come up with opportunities for change, and consider applying a staged set of "treatments" and further interventions.

1.1 What is a healthcare system, and what are its objectives?

We sometimes think of a healthcare system as a collection of clinics, hospital and staff, and business processes that allow this system to treat people when they are sick and keep them well. But many of the health outcomes in a society are a result of a more complex set of interactions within the environment, the society, and between members of the society. In some cases, these latter factors are more important than the way people interface with interventional healthcare systems. It is often said that we need more than the way people interface with interventional healthcare systems. It is often said that we need more health and less healthcare.

Underlying our quest is the natural development, progression, and aging of humans as they lead their lives—physically, mentally, and behaviorally. It is therefore reasonable to characterize a *healthcare enterprise* as consisting of the *set of all actions that individuals, caregivers, healthcare professionals, community, society, institutions, governments, etc., collectively take, or are subjected to, that impact the health outcomes of the population*. All of these entities, whatever their role, impact each other, and so the population evolves. This can be called *population health*.

We start with the simplest and most well-known setting for this evolution, a microcosm, when a patient visits a doctor, sometimes called a *clinical encounter*. After hearing a question or a "complaint," the doctor sizes up the situation, gathers all information at his or her disposal, and makes a choice of what to do. The action chosen (including the choice to just reassure or educate) is based on years of training, established scientific and medical evidence, accepted standards of care, and the presenting state of the patient. Our purpose in this introduction is to characterize this treatment process as a *problem of information* and to examine the process of how we can manage and use information to create actions that move toward better health for the individual as well as the population at large.

Hence, we consider several basic problems confronting us in this task of treating the individual in a healthcare setting. Later, we will consider structural issues with healthcare, the state of scientific knowledge, available services, their payments, etc.

The first setting: when we go to the doctor. We are now at the patient–clinician interface.

Several problems at the patient–clinician interface.

1. Evidence for the most effective treatment may or may not be clear, it may or may not be easily accessible, and treatments may not be uniformly applied.
2. The state of the patient is not always sufficiently evident that the right treatment can be selected or applied.
3. The patient is not always within the reach or influence of the healthcare system at the right times, indeed during vast portions of their lives, for example, their time at home or in the community.

Each of these three problems, and indeed all of healthcare, is a "problem of information." And it is not hard to see why technologies like AI might be poised to address them. The first question above is one of managing and distilling troves of unstructured information. The second may involve summarizing information from many systems, the patient, medical devices, and labs. The third may involve patient attitudes and might be able to be impacted by personalized coaching and nudging.

Starting with the first problem: For many diseases, the evidence base is not always sufficiently developed to know which treatments work and under what circumstances. Evidence generation and curation can, in principle, be addressed by careful examination of a large body of historical scientific evidence (e.g., epidemiology, population health, the medical literature, etc.). This repository of knowledge is constantly being expanded by research, including carefully controlled acts of experimentation (e.g., randomized clinical trials) guided by insights from various life sciences. Lately, information and mathematical sciences are also beginning to play a key role in this process of evidence generation. For example, biostatistics and "big data" techniques can help in processing large amounts of structured and unstructured data. However, mankind's accumulated knowledge resides in a vast and unstructured literature, including the PubMed database [3] that by 2024 has more than 37 million citations and abstracts of articles/papers in the biomedical literature with over one million new articles being added each year (and PubMed is not exhaustive). AI-based natural language processing (NLP), which carries out intelligent search, summarization, consensus, and meta-analysis, is starting to help to deal with this sprawl of information.

It sounds good to be able to have the world's medical knowledge, including analyses and summaries of the literature, at our fingertips using a search tool or chatbot, but things are not as simple as they sound. For example, even though recent papers are published in PDF format, there has not yet been a fully effective method of converting those papers to searchable or ingestible form, especially the tables and figures. Great progress has been made in search, AI and chatbots, but simple stubborn problems remain. Companies are working to chip away at these problems.[1] We are all aware of generative AI's, or large language model's (LLM's), habit of hallucinating made-up facts. Healthcare systems, not wanting to leave things to chance, or suffer too much variation in care, carefully prepare Clinical Practice Guidelines, at least for the most common diseases. While valuable, these can be lengthy and hard to use, and evidence is that they tend not to be consistently followed [4].

Meanwhile AI's latest technologies, including LLMs, are voraciously ingesting significant portions of the published medical knowledge. This is also being done in more systematic and curated ways, such as ClinicalKey AI [5] described later in Chapters 3 and 7. Whole LLMs are being curated specifically for medical knowledge, for example, Med-Gemini.

But it is not just the nitty-gritty of ingesting and interpreting these papers into a large structured or unstructured "model." A more fundamental problem is the fact that a significant portion of medical unstructured "model." A more fundamental problem is the fact that a significant portion of medical studies are not repeatable [6], and there is increasing evidence that quite a few reported experimental studies may be incorrect[2] [7] or may have been superseded by new insights or refinements. Hence, NLP techniques will not only have to regurgitate information but should also *assess, balance, and weigh the correctness of information*, judge validity, quality, and significance of clinical studies and somehow arrive at a *consensus summary* of evidence.

An informative example of the need to carefully assess clinical study results is a 2005 observational study that reported a reduction in cancer incidence among diabetics using *metformin* [8]. This study

[1]E.g., in April 2024, Adobe launched AI Assistant which reads.pdfs. It is powered by Adobe's Liquid Mode, Sensei, and other AI technologies from Adobe.

[2]It is only fair to point out that medical sciences are in a better state than most experimental sciences in that numerous studies are subjected to randomized controlled trials.

generated excitement and resulted in confirmatory studies reporting reductions in cancer incidence and outcomes with metformin use. Tests were conducted in the lab, and theories were advanced to explain how metformin might interfere with cancer pathways.

But then subsequent carefully defined randomized controlled trials found no benefit, as reported as early as 2012 and through 2023 [9]. The authors of the latter paper suggested that a potential flaw in multiple observational studies could be an effect of several time-related biases, including an effect called *immortal time bias*.[3] And that this source of bias, which is precluded by proper randomized trial design, and more careful observational trial designs, such as propensity matching, are well known and should have been used. To quote the authors "many erroneous observational studies affected by time-related biases led to a 17-year quest whereby large, randomized trials found no benefit in cancer outcomes associated with metformin treatment … A November 2022 search of ClinicalTrials.gov on outcomes associated with metformin treatment … A November 2022 search of ClinicalTrials.gov on the terms 'cancer' and 'metformin' revealed around 109 registered ongoing clinical trials that were studying the use of metformin in the treatment of various cancers."

While it is still possible that there could be benefits of metformin in certain cancers, we must conclude that, at best, the evidence is mixed and uncertain. It is going to be challenging to build automated tools that appropriately weigh evidence and scrutinize trial design and statistical methodology, bestow the right kind of skepticisms, and come out with a wise consensus conclusion on what is known and what is not and furthermore what are the correct clinical courses of action. We will follow up in Chapter 7 and ask an AI tool to evaluate this case of uncertain evidence.

The second problem, knowing what actions to take in the presence of a patient, requires sensing the state of the patient through observation, prior records, family and medical history, laboratory tests, current investigations, etc., then comparing this "state and history description" to the large body of medical evidence (the clinician's intrinsic knowledge, treatment guidelines, and reachable documented knowledge), and then making a diagnosis and/or treatment decision. A further problem is understanding and supporting patient preferences, which might not be so amenable to automation and AI.

A clinician must sometimes do all this and make a treatment decision in 7–10 minutes. Inevitably, situations of "cognitive overload" occur—there simply is not time to read and inquire of the patient's history or situation, look at the labs and images, consult the literature and expert colleagues, etc. As a result, operating in the double darkness of not fully understanding the patient and the vast and sometimes inconsistent and unreliable body of medical knowledge, it is not surprising that errors of suboptimal or incorrect treatment, including medication errors, are more prevalent than desired [10]. Clinical practice guidelines are not always followed, sometimes because they do not apply to the case at hand but also sometimes because cognitive information overload does not allow it. Systematic studies such as the RAND Corporation's [4] in the US show that, across a variety of diseases, compliance of clinicians with treatment guidelines ranges through 40%–60% and averages about 55%.[4] Another effect has been characterized as the "know-do gap" [11]. It is hard to get reliable data on the know-do gap; one study [12] found that "… 74% of Indian clinicians were able to tell researchers how to deal with patients

---

[3]Immortal time bias occurs when the study drug, in this case metformin, is started at a later time t than the time s being the start of the comparator treatment. The time from s to t is considered immortal time, since the patient was alive (or cancer-free) during this period, or else they would not have been selected for the study drug, hence biasing the results.

[4]Besides the limitations noted by the authors of the widely cited and quoted paper by McGlynn et al., these further limitations should be noted: the paper from 2003 is now rather dated; it reflects only sample data from the US; the question of how much of the gap is due to clinical versus patient behaviors is not always clear; and the degree of how much deviation from guidelines is acceptable is variable. The limitations of clinical practice guidelines are further discussed in Chapter 8.

suffering from angina, asthma or diarrhea, but when visited by mystery 'patients' presenting with exactly these symptoms, just 31% treated them correctly. One explanation for the 'know-do gap' is that patients generally know far less about the best course of action than clinicians, who can get away with under- or over-treatment when they are not held accountable for their work."

The third problem is more challenging and is related to problems of public health. It is well known that environment, the presence of healthy food choices, safe places to exercise, and a number of other social determinants of health have a significant impact on health outcomes. How do we create a healthy environment, whether it be in the creation of a conducive built environment, health education, green spaces, availability of healthy foods, avoidance of toxins and stressors, poverty avoidance, etc.? Health outcomes are also known to be related to socio-economic status (SES), with lifespans in some countries varying by close to 20 years across SES groups [13]. So, in addition to SES and environmental conditions, lifestyle choices (diet, activity, substance abuse, stress, etc.), which are discussed more below, have an effect that may turn out to be greater than the effects of medical care.

This analysis gives us the insight that will be confirmed with data in Chapter 2 that what happens in the doctor's office is just one event in the whole trajectory of the patient. Both the precursors and consequences of what happens when the patient is at home and in the community suggest the narrative that is taken in Part 3, when we journey through several venues that attempt to influence health outcomes.

But while we are still in the doctor's office, we should be happy to acknowledge that, despite these challenges, manual diagnosis and treatment in the hands of an experienced clinician generally works quite well. Doctors are trained to think in terms of Bayes' rule and render the highest probability diagnosis given the presenting symptoms: "when you hear hoofbeats, think of horses not zebras." And only after doing that to consciously examine differential and less likely diagnoses, known in the trade as "zebras" [14]. A related principle in diagnosis is that of *parsimony*, that the simplest explanation of a cause is often the best one. This principle is sometimes known as Occam's Razor and has a probabilistic interpretation that the cooccurrence of two conditions is usually less likely than one alone. The whole process proceeds in stages: the possibility of finding underlying suspected but unknown conditions ("unobservable states" in systems theory) may require further investigation through lab or imaging techniques, and thereby information may be gathered as underlying conditions or stages of disease progress in a patient. This general complexity of the decision process suggests the use of stage-wise decision support: using data available on the current case, guidance through current clinical practice guidelines, conducting statistical analysis known as *similarity analysis* from a collection of like cases, as well as the ability to reach into the medical literature either in raw or curated form. This is clearly an area where technology and AI may be able to help, and constitutes a general area known as Clinical Decision Support (CDS).

literature either in raw or curated form. This is clearly an area where technology and AI may be able to help, and constitutes a general area known as Clinical Decision Support (CDS).

1.2 What is the "business of healthcare," and why is it important in our transformational quest?

It is a fact of life that a healthcare system needs to have a sustainable business model.[5]

[5]A *business model* is a strategy and plan to create and deliver value, and it may also involve how products or services will be marketed or sold, hence creating value. Some call it the "what" of a business. We will also talk about *business processes*, which are the collection of methods, actions, steps, and procedures that are used to execute the business in a systematic way. They are sometimes called the "how" of the business and include all the workflows that contribute to delivering a product or service. Sometimes, the term *care model* is used in healthcare, and this contains elements of both business model and business process.

Resources are not only finite but are often severely limited or constrained and must be optimally allocated to fairly maximize outcomes under constraints. These issues determine what is sometimes referred to as "the business of healthcare." We will see that business processes and human incentives often have an outsized effect on what treatment decisions are made and how resources are allocated.

It may seem heretical to mention business processes and models when dealing with human life. But the evidence is increasing that costs and the behavioral economics of incentives can be pivotal in getting the right behaviors for both the patient and clinician. Business processes and business models do not only deal with incentives. We shall see that improved business processes can sometimes provide fundamentally different and improved methods, operations, logistics, etc., that are better suited to the problem at hand.

An example of the business processes impacting treatment revolves around questions of over-

An example of the business processes impacting treatment revolves around questions of overtreatment and undertreatment. Although controversial, some evidence (particularly but not exclusively in the US) exists for overtreatment. Welch's book "*Overdiagnosed, Making People Sick in the Pursuit of Health*" [15] provoked considerable discussion when it came out in 2012 [16]. While liability (e.g., practicing of "defensive medicine") could be a major factor in some countries, an arguably more prevalent root cause is simply the lack of clear insight into how a patient should be optimally treated, and therefore a rational approach is to err on the side of helping or acting. But more healthcare is not always better. Nevertheless, referring to the RAND study by McGlynn et al. [4], more instances of undertreatment (or under-use) are reported than overtreatment (over-use). And later in the chapter we will discuss some evidence that overtreatment [17] may *not* be one of the leading causes of high costs in the US health system.

It's not just about diagnosis and treatment. There are sometimes mass campaigns for health screenings before the science has demonstrated benefit. While for some diseases it is likely that screenings help, for other diseases it is possible that screenings do more harm than good. A paper [18] entitled "*Why cancer screening has never been shown to 'save lives'—and what we can do about it*" called for much better evidence generation in the use of health screenings. A particular screening process can be excessive or insufficient and is not necessarily justified purely based on cost effectiveness (i.e., saving more money than it costs). Just like any other investment, understanding that screening budgets are always limited, we should allocate that budget to the screenings that have the best return on investment.

Back to the business of healthcare, in 2009 Clay Christensen teamed with two MDs, Jerome Grossman and Jason Hwang and published the book "The Innovator's Prescription" [19] (which was reissued in 2017). Christensen had earlier achieved attention with his book "The Innovator's Dilemma" where he traced how technological advances often created fertile grounds for industry transformation, and in many cases created disruption in the slate of incumbent industry practices and players. Incumbents often had little incentive to adopt these new technologies. Common patterns of disruption seem to emerge in the chemical industry, vehicles, pharmaceuticals, IT, etc. Why not healthcare?

The Innovator's Prescription focused on business models and on one particularly important aspect: how care is obtained and paid for. The authors argued that *fee-for-service* models are inferior to other models, in part because they may result in overservicing (more generally resource misallocation) and take an overly myopic or transactional view of medical care. The authors cite a quote from the CEO of Kaiser that illustrates the problem. An asthma prevention visit might cost $100, with another $200 being spent for inhalers. But an ER visit would generate $200-$4000 in revenue, and a full hospitalization $10,000 to $40,000. No wonder the revenue-driven systems are not focusing on preventing asthma attacks.

Another business model is known as *value-based care* and could, in contrast, be thought of as *fee-for-outcome*. The influential treatise by Porter and Teisberg [20] argued that focus of care must move to *value* rather than *volume*, with value being defined by the (sometimes incommensurate) quotient of value per unit cost—i.e., emphasis on value of outcomes for any treatment. Some of the pioneering uses of these ideas have been in hip replacement or Coronary Artery Bypass Graft (CABG). A New York Times article [21] reported in 2007 on an innovative care model at Geisinger[6] in Pennsylvania and trumpeted "In Bid for Better Care, Surgery with a Warranty." The genesis of the idea is the *lack of incentive* for best care: "Under the typical system, missing an antibiotic or giving poor instructions when a patient is released from the hospital results in a perverse reward: the chance to bill the patient again if more treatment is necessary. As a result, doctors and hospitals have little incentive to ensure they consistently provide the treatments that medical research has shown to produce the best results."

The Geisinger ProvenCare [22] model also implemented a significant *business process* improvement, taking a page from manufacturing (see Chapter 3). According to Dr. Arnold Milstein, at that time the medical director for the Pacific Business Group on Health, Geisinger "is one of the few systems in the country that is just beginning to understand the lessons of global manufacturing."[7] In reassessing how they perform bypass surgery, Geisinger doctors identified 40 essential steps. Then they devised procedures to ensure the steps would always be followed, regardless of which surgeon or which one of its three hospitals was involved.

Accompanying this is a *business model* innovation: "Geisinger essentially guarantees its workmanship, charging a flat fee that includes 90 days of follow-up treatment. Even if a patient suffers complications or has to come back to the hospital, Geisinger promises not to send the insurer another bill." This incentivizes everyone in the treatment journey to do everything right the first time.

Evidence for the period 2006 - 2015 at Geisinger was subsequently reported [23] for CABG which had also been expanded to multiple conditions, including hip replacement, cataract surgery, etc. Hospital charges decreased 5%. Adherence to evidence-based process measures increased from 59% to 100%. No changes were seen in outcome measures. Other centers trialing bundled care pricing pointed to generally but modestly improved costs.

After introducing business models of Fee-for-service and Fee-for-outcome, Innovator's Prescription introduced a third model handily called *Fee-for-membership*.[8] A prime exponent of this model is Kaiser Permanente. But in this third model, the story gets a little more complicated as it involves the

Kaiser Permanente. But in this third model, the story gets a little more complicated as it involves the double-edged concept of *Capitation*.

Beyond value-based models, the model called Capitation has the potential to work well, but it depends on its implementation. The term "capitation" derives from the term *per capita*, where a fixed fee per head is charged and then care is buffet-style, i.e., bundled into an annual membership payment, or could be free to the patient, or with a small copayment. If the payer or insurer is integrated with a

[6]Geisinger was acquired by Risant Health, a subsidiary of Kaiser Permanente in April 2024.
[7]See the annex of Chapter 3 for comparisons of healthcare with manufacturing.
[8]In "The Innovator's Dilemma," Chapter 4, the authors draw a one to one relationship between solution shops (fee-for-service), value adding process (fee-for value), and disease management networks (fee-for-outcome). However, a conversation (July 9, 2024) with the surviving author Dr. Jason Hwang makes clear that this one to one relationship has frayed in intervening years. New instances of providers have arisen with varying payment models combined with these three different delivery models. Experimentation with medical business models continues unabated.

provider, and patient turnover is not too great, then the provider has the incentive to keep the patient and family (if included) well. These are called *integrated capitation plans*. When an insurer is focused on selling policies and minimizing claims, and client turnover is high, or the insurer/payer is buying services from a variety of providers, i.e., *non-integrated capitation*, then there is an incentive for the insurer to become a gatekeeper that limits care, and there is a downward path to restrict care. We may start to see "lemon dropping" where needy patients are dropped and or "cherry picking" when well patients are preferentially recruited.[9] Integrated capitation, done well, encourages appropriate behaviors at the clinician-patient interface and places emphasis on prevention and patient lifestyles. Such an "integrated fee-for-membership" business model can place far more emphasis on wellness and prevention and should have less tendency to undertreat or overtreat. A systematic review [24] showed that "there was some evidence that primary care physicians provide a greater quantity of primary care that "there was some evidence that primary care physicians provide a greater quantity of primary care services under fee for service payment compared with capitation and salary, although long-term effects are unclear. There was no evidence, however, concerning other important outcomes such as patient health status, or comparing the relative impact of salary versus capitation payment." However, it was also noted that the generalizability of these conclusions is unknown. Non-integrated capitation may also work better when it follows the principles of an Accountable Care Organization (ACO). ACOs are a concept promoted by the US government since 2011 by which multiple medical services are co-ordinated, with the goal of providing "accountable" healthcare, involving both quality and costs.

Besides these three models of payments, Christensen et al. also made significant recommendations about how healthcare operations and the business models of existing healthcare companies could be refined by being "teased apart." They considered how existing healthcare enterprises could be decomposed into species that would take a particular approach to the way they work. One "business model species" is termed a "Solutions Shop"[10]. It provides a complete solution to a relatively complex process, e.g., the diagnosis of a condition possibly using advanced methods such as imaging, advanced diagnostics, etc. These often operate as fee-for service, since it is a priori unclear what extent of services may be required. Fee-for-service makes sense since the provider doesn't know in advance what they are going to find, and they may have a more transactional than long-term relationship with the patient. Examples in the US are highly skilled solution providers such as Cleveland Clinic, Mayo Clinic, Memorial Sloan Kettering Cancer Center, MD Anderson Cancer Center, etc., to which many patients will travel for evaluation, second opinion or treatment for serious disease.

The second class of models is called "Value-Added Processes" (VAPs) and mostly rely on relatively standardized inputs and goals. The idea of Value-Added Processes is that they have a very clear delivery goal (referred to as "a job to be done"). They also often have specific resources to support this standardization, for example standard protocols, process lists, etc. Patients go there when they know what they want done, and benefit from the high quality that results from standardized and rigorous processes (similar to what is found in Toyota's famous manufacturing system). An example is CVS's retail Minute-Clinics which carry out very well-defined functions, e.g., a respiratory consultation and treatment, and they also have fixed fees, which may be bundled payments. A hospital example would be a hip replacement or CABG, such as described above as pioneered by systems such as Geisinger.

---

[9]Note that cherry picking can be addressed using the risk models discussed in Chapter 8 to set risk-adjusted payments.

[10]The three models that Innovator's Prescription describes stem from earlier work that defines three types of job-focused business models: "chains," "shops," and "networks." Stabell, Charles B., and Øystein D. Fjeldstad, "Configuring value for competitive advantage: on chains, shops, and networks." *Strategic management journal* 19.5 (1998): 413–437.

Perhaps the best example of a VAP is Teladoc [25], a large telehealth outfit active which is stated to be active in 130 countries, serving 80 million people. It claims to resolve 92% of issues in one visit and uses over 100 proprietary clinical guidelines. Nowadays a considerable portion of Teladoc's revenue comes from mental health services after it acquired betterhelp.com.

The Innovator's Prescription recommended that hospitals break their operations into these two distinct models, solution shops and VAPs, and adopt business, quality and staffing models accordingly. Indeed, Teladoc claims alliances with 60 hospitals, suggesting that this decomposition of the business model may in fact be taking place.

The third class of models is called "Facilitated Networks." These networks focus on enabling and coordinating care. They allow users and contributors to manage their overall health within a network, to help each other and share components of solutions.[11] These services have their origins in patient support groups, a well-known successful model, and they have moved online, especially driven by the pandemic. Well known examples are Alcoholics Anonymous, PatientsLikeMe [26], Smart Patients [27] for cancer, and Let's Talk [28] for mental health. Some of these services, paid or unpaid, have had success and have significant enrolments. Other commercial examples include Zocdoc [29] that lets patients find clinical services and get appointments; additionally, they carried out many vaccinations during the pandemic. At an even higher level of expertise, Grand Rounds provides access to specialists for second opinions and continues to expand with other mergers [30].

These service navigators, networks and specialty companies (sometimes provided as an employee benefit) fulfill part of the need of navigation but still leave serious gaps and fail to serve many needs in the community. Some hospitals have also adopted them as an efficient way of expanding their reach and as patient recruitment tools.

The fact is that for a patient confronting a serious illness, or even a chronic disease, it is hard to get information and then navigate the medical system. Some patients are lucky enough to have a family member who can help coordinate their appointments, their medications, etc. But many patients with comorbidities have multiple doctors and do not always receive consistent or understandable advice or services. The hope would be that their primary care physician (PCP) can help to provide this kind of coordination. But given busy schedules this is usually not possible or manageable, especially with the level of payments or reimbursements that PCPs receive. Sometimes an integrated provider (such as

[11] Innovator's Prescription poses three business models for healthcare: Solution shops, value-added processes, and facilitated networks. It is worth mapping them to well established principles in business strategy. Porter, in his well-known teachings

networks. It is worth mapping them to well established principles in business strategy. Porter, in his well-known teachings https://en.wikipedia.org/wiki/Porter%27s_generic_strategies, stated that business should adopt one of three strategies: Cost leadership, differentiation, or focus. The idea is that you should only pick one of these to succeed. Tracey and Wiersma, https://hbr.org/1993/01/customer-intimacy-and-other-value-disciplines, also promote a tryptic of strategies, which are called operational excellence, product leadership, and customer intimacy. They also state that you should pick only one. In Innovator's Prescription, the idea of a value-added processor with its standard repeatable process has elements of cost leadership and focus and operational excellence. The idea of a solution shop with its distinguished high level of often-branded expertise has elements of differentiation and product leadership. The last one, facilitated networks, is a more unusual and nonproduct concept. It fills the need for an antidote to complexity in navigating healthcare systems. When successful, it has elements of customer intimacy and, as a concierge service, has elements of product differentiation. As we trace the development of instances within these three models, you will see that they are often combined into one provider service to serve different needs in a patient's life journey. As the patient confronts both simple and life-threatening diseases, they find complexity and unknowns. The story is an overwhelming culmination of patients finding their way through a complex health system while, in parallel, healthcare providers, insurers, etc., are working to optimize revenue, profit, and competitive goals, sometimes in the context of a set of constraining values.

Kaiser Permanente) will provide this service and have processes in place to remind a patient to have a colonoscopy, get a flu shot or a screening, etc. But when it comes to serious disease, patients can easily become overwhelmed with the extensive coordination required. In some cases, insurance companies have tried to provide this kind of facilitation by hiring nurses as patient advisers, but their calls are often met with suspicion from patients who are not sure that the insurance company is really trying to help them or is just working to minimize claims.

In the US, concierge medical services have sprung up [31] and while they may provide some simplification of care access for well-heeled consumers, these premium services can further exacerbate inequities [32]. Doctors who used to treat 2000 patients are now providing better service to just 500 fee-paying patients, and with doctor shortages, this leaves other patients underserved and causes increases in waiting times, if they can get care at all.

fee-paying patients, and with doctor shortages, this leaves other patients underserved and causes increases in waiting times, if they can get care at all.

The author managed to catch up with Dr. Jason Hwang, the only surviving author of Innovator's Prescription, in July 2024. While many of the innovations predicted in the book are playing out, in terms of payment models the status quo of fee-for-service has tended to prevail.

According to Dr. Hwang, perhaps the slowest model to take hold and penetrate has been the Facilitated Networks. As described, there are instances of the latter services being provided to employees, and they are sometimes improvized by digital natives, but most members of the public still do not have access to these kinds of services, even though some of them are free (e.g., Smart Patients). While many patients will google a question about healthcare, they are slower to sign up for a group, perhaps because of trust factors and fears about data security and privacy. It is interesting to think about how GenAI could help with this need, in the same way that it is helping to map out a vacation which also sometimes involves an overwhelming set of choices, but for which free data (not always reliable) abounds through services such as Tripadvisor.

Dr. Hwang also highlights that while these focused business models are forming, sometimes as a pure-play or in combinations, consolidation is also a major force. Large health groups, insurance companies, pharmacy chains and tech companies are buying up private practices and using their size and reach to consolidate the landscape [33]. For example, Amazon reentered health by acquiring One Medical in 2023, after the abortive effort of Haven Health described in the Introduction.

A lot of concerns are being raised with these new entities, particularly regarding the autonomy of doctors to provide quality care, e.g., Ref. [34]. In the US, all over-65s can get federal government-provided health insurance through the government-provided Medicare program. Alternatively, over-65s may choose Medicare Advantage plans [35], which are offered by insurance companies, who offer different programs, sometimes with less copay but sometimes with more restrictions and limitations. Medicare Advantage programs can be profitable to insurance companies, although there are many complaints by the customers concerning denials and limitations of care. Medical groups, especially those in primary care, frequently report feeling disempowered and overloaded with obtaining authorizations after being acquired by these mega-companies and workers sometimes suffer significant burnout. One result is that fewer medical graduates are choosing a primary care career, further contributing to shortages of family doctors. The impact of these acquisitions remains to be seen, and the US healthcare system remains highly fragmented. Patients over 65 can usually obtain care through Medicare programs, and the fully employed and the well-heeled get good access to comprehensive plans and insurance that blunts the impact of any major illness. Obamacare, or the Accountable Care Act, has helped millions of people gain access to health insurance. But as of 2022 [36], 10.2% of people under 65 in the US remain uninsured.

A final comment on the larger healthcare systems and private equity players buying up medical practices and either incorporating or restructuring them into new corporate structures and business models: while some of these activities are attracting a lot of criticism, the introduction of new business models could in principle be a force for good. The introduction of professional business processes and models into healthcare can bring welcome innovation and valuable changes, as we shall see in the rest of Part 1. It can also introduce new efficiency in operations and marketing of medical services and improve consumers' awareness of health choices. This book focusses on the opportunity for technology and AI to help in making these improvements and so transform healthcare. Adopting a professional business approach, focused on outcomes as well as costs and efficiency, which will include deploying the best and most practical forms of technology and AI, could be a positive move toward adopting needed changes and creating new ways of improving care.

We sum up on the matter of costs and their trends. It is often repeated that without intervention, healthcare will eventually bankrupt every country in the world. A welcome but economically disruptive trend of much longer lifespans has been brought about by the miracle of health and life sciences. One result is that chronic diseases of diabetes, hypertension and dyslipidemia have now replaced infectious diseases and trauma as the highest sources of burden of disease in most countries.

While most economic and engineering systems exhibit a quality-to-cost tradeoff, healthcare systems are often presently "so far off the efficient curve" of economics that it is conceivable that, simultaneously, care could be both improved and costs lowered—if the population was kept constant. However, the size and demographics of the population is not constant and longer lives and greater incidence of chronic diseases means that quality and effectiveness must be improved while keeping costs in check.

As we have seen, this requires the judicious selection of new business models and deployment of new and integrated healthcare processes. There is great hope that their success can be aided by careful and selective application of technology. Every one of these proposed innovations depends on information availability, how resources are identified and allocated, how providers operate, or how patients receive care. We will first ask what pain points must be addressed; this will help identify key opportunities for improvements. Only then, we ask what transformations and new perspectives are needed. And only after that do we ask what technologies are needed to support those transformations.

1.3 Lessons in transformation from various countries around the world

1.3 Lessons in transformation from various countries around the world

Healthcare systems around the world have arisen relatively independently and have undergone optimization driven by patient demand, political forces (the expression of and attempts to address these demands), cost pressures, or perhaps some random chance or luck in how the system was first formed. It is interesting to enquire whether these somewhat different "experiments" might yield stories and comparators about healthcare systems, describe alternative approaches and transformations, and disclose what knobs can be turned to improve and optimize outcomes.

Whichever medical system that we discuss, you can be sure that there will be multiple opinions and theories about what ails it. Some countries' systems have essentially removed the problem of the patient's ability to pay to obtain care by providing a form of "universal care" [37], but then they often receive criticism for long waiting lines for treatments. The United States (US) may offer some of the best treatments in the world but not all can afford the treatments, and some may even be bankrupted by

disease or die because of lack of appropriate care. Given all the confounding variables, it is not easy to systematically compare healthcare systems. Nevertheless, some comparisons are informative. And various systems have characteristics that suggest that ideas could be borrowed between systems.

Why do health systems in some countries have such different costs and outcomes? In considering comparisons between nations, we have already described several trends and business models within the United States. According to the Centers for Medicare & Medicaid Services (CMS) [38], "U.S. health care spending grew 7.5% in 2023, reaching $4.9 trillion or $14,570 per person. As a share of the nation's Gross Domestic Product, health spending accounted for 17.6%." This was a 0.3% increase as a portion of GDP over 2022.

This massive spend in the US does not result in stellar outcomes, and, in comparison, European systems spend in the range of 10%–12% of GDP. Singapore, for example, spends a much lower portion systems spend in the range of 10%–12% of GDP. Singapore, for example, spends a much lower portion of GDP with some of the world's best outcomes [39]. Bear in mind that comparing various systems is not necessarily apples-to-apples because nations have differing populations (including different age distributions and different genotypes), different challenges and instigators of disease (e.g., prevalence of smoking or other addictions), different cost structures in their healthcare systems, and different levels of affordability within their national budgets.

It is common to attribute the high costs in the US as coming from defensive medicine, over-servicing, overdiagnosis, and overuse. While some overdiagnosis may exist, as discussed above and articulated by Welch, many will be surprised to learn that utilization does not appear to be a prime driver of higher costs in the US.

A careful analysis in JAMA by Papanicolas, Woskie, and Jha in 2018 [17] analyzed spending parameters in depth, comparing the US with 10 other high-income countries, and gives considerable insight. It finds that the workforce numbers of doctors and nurses in the US are not that different from other countries nor are utilizations of hospital beds, although US utilization of CT and MRI studies was about 1.5 times higher than the mean elsewhere. Instead, US healthcare's relatively larger costs are primarily attributed to:

a. Higher clinician salaries. For example, generalist physician salaries are 1.5–3 times that in comparator countries.
b. High costs of drugs. It is well-publicized that the same drug is much more expensive in the US than in Canada, Europe, or Asia. This is partially caused by the fact that the US Medicare system, the essentially public (although it has private supplements) insurance system for over-65s, has not been allowed to negotiate with drug companies. This is obviously a contributing factor since such a high proportion of medical costs come from seniors. Pharmaceutical costs in the US are 1.5–3 times the costs in other countries.
c. Higher costs of administration: 8% in the US versus 1%–3% in other countries. This is in turn caused by the fragmentation and complexity of the various insurance, payments, and delivery systems in the US.

A great deal of insight in comparing systems from around the world can be found in the remarkable small volume "*In Search of the Perfect Health System*," by Mark Britnell [40]. It reports on about 30 countries, with comparative data and records of success.

What is interesting about these descriptions is that no one system stands out as the best, but several have dimensions of excellence that we can learn from. Britnell cites the universal healthcare of the UK, primary care in Israel, mental health services in Australia, health promotion in the Nordic countries, patient and community empowerment in Africa, innovation flair and speed in India, and aged care in Japan, among others. As one goes through this book and reads about each country, we find that these various levers have been pulled to good effect, but few with total transformative or step function impact. So far there seems to be no "golden screw" that makes everything hold together and singularly creates excellence.

Selecting from Britnell's country-by-country analysis, we selected just four examples and dug in further to see if they might hold lessons for transformation. While Britnell pinpointed more countries with interesting successes, in researching these countries, the author found varying amounts of current evidence, and so we will report only on those for which he was able to find the strongest cases.

1. **Israel seems to have mastered primary care**: The author gained the same impression when he visited their Clalit and Maccabi Health Maintenance Organizations (HMOs) and their Ministry of Health in 2017. Israel's numbers seem to bear it out, with one of the longest lifespans at 82.1 years and a health spend of GDP at 7.2%. Israel manages with one of the lowest numbers of acute care beds in the world. Technology pervades their systems, and they make extensive use of AI, as the author observed during the COVID-19 pandemic when Clalit described[12] a clever program to prioritize care by patient and reach out to them in the order of a risk-prioritized list during the early days of the pandemic. Clalit does a lot of telecare and services via mobile phones and has a strong EHR system. The tech-centered nature of Israeli innovation is well-entrenched in their healthcare systems. Their primary care centers focus on preventative services, but they also house quite a few specialist doctors and integrate them into the practice, so patients do not have to go elsewhere for secondary services. This experience and success suggest that if you can shift from hospitals into clinics and from clinics into the home (called shift-left), there may be great opportunity for transformation—we will see this in the next chapter. Of course, no system is perfect, and it is not surprising that citizens of Israel express more satisfaction with primary care than with hospital services. We will describe further AI-based innovation from Israel in Chapter 8. **Lesson:** There is value in doing more in primary care, the so-called shift-left and shift-down that is described in the next chapter.
2. **Mental health is a successful focus in Australia.** As the reader will see later in the book, we focus a lot on mental health because it tends to be under-resourced and even neglected in many countries and is a fertile area for AI. Looking at various Institute for Health Metrics and Evaluation (IHME) reports [41], we see that mental health often exceeds the burdens of disease (especially in years lived with disabilities) of cancer and cardiovascular disease. This is partly

Evaluation (IHME) reports [41], we see that mental health often exceeds the burdens of disease (especially in years lived with disabilities) of cancer and cardiovascular disease. This is partly because it disables people at all stages of life rather, whereas cancer and cardiovascular diseases are much more prevalent in middle and later life. As we deploy mental health tech in Singapore, we often find that Australia has the models that we want to emulate, whether through Headspace, Black Dog, MindSpot, etc. Australia focused early on moving patients with serious mental illness out from live-in facilities back to the community and was also a pioneer in early psychosis intervention [42]. **Lesson:** Focus on where burden of disease suggests that there will be highly cost-effective programs.

3. **Patient and community empowerment in Africa:** While Britnell highlights patient empowerment in Africa (and goes into more detail on South African healthcare), the author's

[12]Personal communication with Dr. Noa Dagan, Clalit researcher, Tel Aviv, 2020.

experience in poorer parts of Africa is that when patient empowerment is combined with mobile health, a rapid filling of a vacuum occurs. It is reported, in a systematic review [43], that mHealth (use of mobile devices to support health) has reached over 60% penetration in sub-Saharan Africa, although it is difficult to establish an exact current penetration and degree of effectiveness. Another systematic review [44] draws a similar conclusion that there is still insufficient evidence of effectiveness of these techniques such as mHealth, with many of the efforts being characterized as "pilotitis." However, a subsequent and careful 2016 analysis [45] by Eva FM Krah and Johannes G de Kruijf gives hope. In contrast to the reservations of prior systematic reviews and despite the failure to sometimes adhere to Western standards of evidence, this article reports that quite a number of projects have had significant or generally positive results. Further details are as follows:

follows:

a. Patient follow-up and medication adherence: a good portion of trials show feasibility and generally positive and significant results
b. Communication and information for healthcare workers: some positive results with evidence of significant change but several posing concerns
c. Health promotion and disease prevention: encouraging results especially in HIV.

Many projects in Africa suffer from poor infrastructure, but SMS texting can be a viable and sufficient vehicle for mHealth. Confidentiality and privacy concerns are prevalent, with access being limited by the power of male family members and cultural beliefs and expectations. Overall, 71% of 65 projects reviewed resulted in a generally positive project evaluation. The authors call for a "constructive realism" to guide progress in Africa. Again, success often requires participation of local private service providers and collaboration with local partners and a user-centric design to facilitate adoption. **Lessons:** When applying technology, remember the principle of "good enough" and that a pure tech push will not overcome cultural and access limitations. Also, we must be careful applying purist and western standards of high evidence to impede the promotion of promising projects that fill a vacuum—the alternative may be no care at all. Unfortunately, funding remains insufficient and a solid argument for building a greater evidence base is that it will heighten justifications for investment in scale-up mHealth projects. Tech alone will not solve what is an enormous unmet need in developing countries, and it must still be combined with human-based services where needed. Some of these principles are further expanded upon in the Chapter 13, "How AI can transform care in emerging economies."

4. **Aged care in Japan**: There are many reports of aged care being exemplary in Japan [46,47]. Japan is of course a superaging society with a high aged dependency ratio (the ratio of those over age 65 years to the working age population) compared to other developed countries. Affordability is managed via citizen contributions from age 40 years and a 10% individual copayment at the time of accessing services. Aged care facilities have a concentration on nutritional support (largely traditional and plant based) and a focus on recovery, rehabilitation, and maintenance of functional and cognitive capacity. The Japanese traditions of observing seasonality with artforms and experiencing of festivals and plant life provide additional cognitive stimulation. Older people are specially accommodated in public transit systems and considered assets to the society and encouraged to share their skills in cooking and growing vegetables. Lifespan after admission to aged care is 44% longer in Japan versus US or Australia, although causality for Japan's larger figure is not evident. Moreover, only 6% of over-65s live in aged care facilities, and a lot of seniors are also cared for by seniors (in 63.5% of the households in Japan where a person 65 years

or older is receiving care, the caregiver is also a senior, at least 65 years' old [48]). In a careful comparison of aged care in Japan versus UK [49], the authors note that Japan has a more national (with uniform quality) and financially provided-for model and a greater commitment to deployment of technology (assistive robots, sensors, and artificial intelligence), whereas shortage of skills and required proofs of cost-saving may hold up these deployments in the UK. **Lesson:** A focus on the quality and location of aged care has high leverage, and strategies based on effective funding programs, innovative workforce resourcing, and technology have high potential.

Having looked at these examples by country, it is informative to understand how Britnell wraps up his comparative-by-country examples and to see which transformative mechanisms, or best practices, he highlights. This will give some hints and previews as we investigate technology-based interventions in future chapters.

1. **Which funding system is best?** Universal healthcare is often taken as meaning that all a country's residents have access to healthcare that they can afford. This includes healthcare promotion and prevention, treatment and recovery from disease, rehabilitation, and palliative care. Defined in this way, universal healthcare should be a global aspiration. However, only about 40% of the world's population has universal healthcare, and one of the world's richest countries, the US, does not. Families in the US can easily be bankrupted by an illness or be unable to afford the healthcare that they need that could save their lives. Across the Organisation for Economic Co-operation and Development (OECD), funding sources average 72% for public contributions and 28% for private. Data for current portions of GDP in universal healthcare systems implemented in Asia (Japan, Singapore, Korea, and Hong Kong) exhibit countries with leading life expectancies at costs of 5%–9% of GDP. Good insurance-based systems such as those of Switzerland, Germany, France, and the Netherlands also yield good results at 11%–12% of GDP. The US is of course an outlier at 17%; some of the key causes of this high figure were explained above. Systems that have single pricing systems and a dominant player such as in the Nordics, Japan, and the UK deliver life expectancies of 80+ with about 9%–11% of GDP. These countries have simpler and quite efficient systems, in part because of reduced fragmentation, lower cost of administration and premiums, etc., although it is sometimes suggested that they may have less competition and therefore adopt innovations more slowly.
2. **In terms of transformative innovations:** All kinds of healthcare systems can harbor interesting transformations and business models; a prime example is the very effective Kaiser Permanente in

transformations and business models; a prime example is the very effective Kaiser Permanente in the US. Nevertheless, fee-for-service is still widespread and can be insidious, resulting in inequities of care and relatively inefficient systems that fail to focus on prevention.

3. **Genomics and personalized medicine:** Not long after the announcement of the sequencing of the human genome, it was suggested that the genome would allow us to have *precision* medicine in addition to *personalized* medicine. Many would say that we should always have personalized medicine, with or without the genome. While genomics has high potential and an important part to play, it is now considered one of a much broader inventory of techniques that we presently have at our disposal. And lately, the term personalized medicine is reappearing as an emphasis, reinvigorated by many other new tools. Medicine should always be delivered with precision and be accurate and precise, and evidence-based treatments should be based on precise knowledge of the best care and the detailed state of the patient. Doing so, with all the data it entails, can be helped by new algorithms and advanced AI techniques. Treatments will then be appropriately

personalized and applied in a way that suits the particularities of the patient: their gender, age, or their way of learning, adapting, and integrating healthcare with their own unique lifestyle and preferences. If possible, care should still be delivered by a person, who can exhibit empathy and caring.

4. **Mobile health:** We have just seen how mobile health in Africa is blazing a new path and has moved from a state of many pilots to what is now several quite convincing systems while still underfunded and needing to be scaled up and further deployed. The mobile phone has become the *universal sensor*, as we describe in Part 3, and also the *universal actuator*, meaning that it can be there at the right time to help manage a behavior change or connect with a clinician. We will explain how mobile devices can be used to change the venue of care in Part 3, and how they can become a prime delivery vehicle for AI-based interventions.

become a prime delivery vehicle for AI-based interventions.

5. **Wearable technology:** While a smartphone can detect many useful biomarkers, there are some measurements that are better served by a wrist or ring device, such as accurate methods of measuring sleep, heart rate, heart rate variability, respiration, and temperature. More advanced wearables are now able to detect blood pressure and blood sugar levels. These supplement the mobile phone for both health promotion and recovery.
6. **More care at home:** As we will spell out in the next chapter, the home and the community are important places to optimize healthy behaviors. They are also appropriate for interventions at all life stages, including aging and palliative care. Mobile health and AI technologies can deliver both automated nudges as well as the teleported personal touch of a care team. This will be a central theme of Part 3.

1.4 Other considerations in transforming healthcare

Having discussed fundamental process and business transformations, and instructive examples from around the world, there are several other trends that should be mentioned. While some of these are not directly related to technology, they should be kept in mind as being important opportunities for transformation.

1. **Better use of team skills in primary and specialist care:** New clinical team models for healthcare are showing success. All members of the healthcare professional team, doctors, nurses, physicians' assistants, pharmacists, physical and occupational therapists, dieticians, psychologists, counselors, and social workers have a critical role to play in caring for and activating patients. Novel substitutional or combinational deployments of team members are now being employed that show improved outcomes, especially in achieving better adherence to therapies and changes in lifestyle factors. Some examples of these novel models include frequent remote consultation and medication adjustments led by nurse practitioners (versus cardiologist visits) in post-AMI care (showing noninferior outcomes) [50]; creating integrated practice units consisting of nonphysician personnel who are dedicated to a specific disease condition for the full cycle of care [51]; and use of teamlets [52], which are a small teams of several doctors and nurses, and a care coordinator, who can maintain continuity of contact with assigned patients in a high volume primary care setting (with results including better screening compliance and fewer chronic disease ER visits).

2. **Other and better methods of patient activation:** In the next chapter, we will go into "the 40/30/20/10 rule," which shows that patient activation and lifestyle alterations are some of the biggest levers especially in chronic diseases. The physician often does not have enough time to spend on detailed counseling for these changes and when he or she does they are often poorly adhered to. The broadening of the care team can also be effectively extended with support groups, which are widely used and were extended to social networks during the COVID-19 pandemic. Community health workers and lay counselors are also beginning to be trained and deployed with good effect, and there are reasons to believe that they can sometimes better relate to the patient. Peer support using digital platforms and social networks is further explored in Chapter 10.
3. **Public health initiatives ranging from awareness campaigns to improvements in health financing and subsidies:** Unmet needs in healthcare have a wide set of causes and remedies that involve awareness, costs, waiting times, distance to care settings, etc. Educational initiatives, particularly around lifestyle and chronic diseases, are effective and essential in public health. Health financing and subsidies vary greatly around the world, with many individuals still being unable to afford care and medications, and some just unable to navigate overly complex systems or failing to understand their needs and importance [53–55]. Other industries have switched their marketing spend to digital with more cost-effective and targeted results, and the potential seems high for healthcare to do the same.

As care becomes vested in these more plural and team-based settings and complexity and information overload increases, the need for collaborative care increases across many professions. And there are now more diverse ways to effectively and efficiently reach and engage patients. It does not take much imagination to see the role that technology and AI might be able to play as an assistive medium. While much of this opportunity is still to be realized, previews will be seen throughout the rest of the book.

References

[1] S. Berg, Family Doctors Spend 86 Minutes of 'Pajama Time' with EHRs Nightly, Available: https://www.ama-assn.org/practice-management/digital/family-doctors-spend-86-minutes-pajama-time-ehrs-nightly. (Accessed 11 January 2025).

[2] S. Berg, Burnout on the Way Down, But 'Pajama Time' Stands Still, Available: https://www.ama-assn.org/practice-management/physician-health/burnout-way-down-pajama-time-stands-still. (Accessed 11 January

[2] S. Berg, Burnout on the Way Down, But 'Pajama Time' Stands Still, Available: https://www.ama-assn.org/practice-management/physician-health/burnout-way-down-pajama-time-stands-still. (Accessed 11 January 2025).
[3] PubMed—National Library of Medicine, Available: https://pubmed.ncbi.nlm.nih.gov/. (Accessed 11 January 2025).
[4] E.A. McGlynn, et al., The quality of health care delivered to adults in the United States, N. Engl. J. Med. 348 (26) (Jun. 2003) 2635–2645, https://doi.org/10.1056/NEJMsa022615.
[5] World Class Clinical Information Meets Artificial Intelligence, Available: https://www.elsevier.com/products/clinicalkey/clinicalkey-ai. (Accessed 11 January 2025).
[6] How Science Goes Wrong, Available: https://www.economist.com/leaders/2013/10/21/how-science-goes-wrong. (Accessed 11 January 2025).
[7] J.P.A. Ioannidis, Why most published research findings are false, PLoS Med. 2 (8) (Aug. 2005) e124, https://doi.org/10.1371/journal.pmed.0020124.

[8] J.M.M. Evans, L.A. Donnelly, A.M. Emslie-Smith, D.R. Alessi, A.D. Morris, Metformin and reduced risk of cancer in diabetic patients, BMJ 330 (7503) (Jun. 2005) 1304–1305, https://doi.org/10.1136/bmj.38415.708634.F7.
[9] O.H.Y. Yu, S. Suissa, Metformin and cancer: solutions to a real-world evidence failure, Diabetes Care 46 (5) (May 2023) 904–912, https://doi.org/10.2337/dci22-0047.
[10] M.S. Donaldson, J.M. Corrigan, L.T. Kohn, To Err Is Human: Building a Safer Health System, 2000.
[11] In Poor Countries It Is Easier Than Ever to See a Medic, Available: https://www.economist.com/international/2017/08/24/in-poor-countries-it-is-easier-than-ever-to-see-a-medic. (Accessed 11 January 2025).
[12] The Importance of Primary Care, Available: https://www.economist.com/special-report/2018/04/26/the-importance-of-primary-care. (Accessed 11 January 2025).
[13] Britain has Endured a Decade of Early Deaths. Why?, Available: https://www.economist.com/leaders/2023/

[12] The Importance of Primary Care, Available: https://www.economist.com/special-report/2018/04/26/the-importance-of-primary-care. (Accessed 11 January 2025).
[13] Britain has Endured a Decade of Early Deaths. Why?, Available: https://www.economist.com/leaders/2023/03/09/britain-has-endured-a-decade-of-early-deaths-why. (Accessed 11 January 2025).
[14] J. Groopman, How doctors think, Houghton Mifflin Co., Boston (2007).
[15] H. Welch, L. Schwartz, S. Woloshin, Overdiagnosed: Making People Sick in the Pursuit of Health, Beacon Press, 2011.
[16] L. Lichtenfeld, Overdiagnosed, J. Clin. Investig. 121 (8) (Aug. 2011), https://doi.org/10.1172/JCI57171.
[17] I. Papanicolas, L.R. Woskie, A.K. Jha, Health care spending in the United States and other high-income countries, JAMA 319 (10) (Jan. 2018) 1024, https://doi.org/10.1001/jama.2018.1150.
[18] V. Prasad, J. Lenzer, D.H. Newman, Why cancer screening has never been shown to 'save lives'—and what we can do about it, BMJ (Jan. 2016) h6080, https://doi.org/10.1136/bmj.h6080.
[19] C.M. Christensen, J.H. Grossman, J. Hwang, The Innovator's Prescription, McGraw Hill, 2009.
[20] M.E. Porter, Redefining Health Care : Creating Value-Based Competition on Results, Harvard Business School Press, Boston, Mass., 2006.
[21] In Bid for Better Care, Surgery With a Warranty—The New York Times, Available: https://www.nytimes.com/2007/05/17/business/17quality.html. (Accessed 12 January 2025).
[22] M. M. Interview with Ronald A. Paulus, ProvenCare: Geisinger's model for care transformation through innovative clinical initiatives and value creation, Am Health Drug Benefits 2 (3) (Apr. 2009) 122–127.
[23] T. Shih, L.M. Chen, B.K. Nallamothu, Will bundled payments change health care? Circulation 131 (24) (Jun. 2015) 2151–2158, https://doi.org/10.1161/CIRCULATIONAHA.114.010393.
[24] T. Gosden, et al., Capitation, salary, fee-for-service and mixed systems of payment: effects on the behaviour of primary care physicians, Cochrane Database Syst. Rev. 2011 (10) (Jul. 2000), https://doi.org/10.1002/14651858.CD002215/ABSTRACT.
[25] Teladoc Health—Wikipedia, Available: https://en.wikipedia.org/wiki/Teladoc_Health. (Accessed 11 January 2025).
[26] Live Better, Together! | PatientsLikeMe, Available: https://www.patientslikeme.com/.
[27] Smart Patients An Online Community for Patients and Their Families, Available: https://www.smartpatients.com/. (Accessed 11 January 2025).
[28] It's Ok to Talk About Your Feelings, Available: https://letstalk.mindline.sg/. (Accessed 11 January 2025).
[29] Zocdoc—Wikipedia, Available: https://en.wikipedia.org/wiki/Zocdoc. (Accessed 11 January 2025).
[30] H. Landi, Doctor On Demand, Grand Rounds Combine to Form Multibillion-Dollar Digital Health Company, Available: https://www.fiercehealthcare.com/tech/doctor-demand-grand-rounds-combine-to-form-multibillion-dollar-digital-health-company. (Accessed 11 January 2025).
[31] D. Robinson-Walker, What is Concierge Medicine and Is It Worth the Price Tag?, Available: https://www.forbes.com/health/healthy-aging/concierge-medicine/. (Accessed 11 January 2025).
[32] Many Doctors are Switching to Concierge Medicine, Exacerbating Physician Shortages, Available: https://www.scientificamerican.com/article/many-doctors-are-switching-to-concierge-medicine-exacerbating-physician-shortages/. (Accessed 06 January 2025).

[33] Corporate Giants Buy Up Primary Care Practices at Rapid Pace, Available: https://www.nytimes.com/2023/05/08/health/primary-care-doctors-consolidation.html. (Accessed 06 January 2025).
[34] Changes at Amazon-Owned Health Services Cause Alarm Among Patients, Employees, Available: https://www.washingtonpost.com/technology/2024/02/28/amazon-health-one-medical/. (Accessed 06 January 2025).
[35] Compare Original Medicare and Medicare Advantage, Available: https://www.medicare.gov/basics/get-started-with-medicare/get-more-coverage/your-coverage-options/compare-original-medicare-medicare-advantage. (Accessed 06 January 2025).
[36] National Center for Health Statistics, U.S. Department of Health and Human Services, Centers for Disease Control and Prevention, Number 194, November 9, 2023.
[37] The Future of Health System, Available: https://www.oecd.org/en/topics/policy-issues/the-future-of-health-systems.html. (Accessed 06 January 2025).
[38] National Health Expenditure Data, Available: https://www.cms.gov/data-research/statistics-trends-and-reports/national-health-expenditure-data. (Accessed 06 January 2025).
[39] C.C. Tan, C.S.P. Lam, D.B. Matchar, Y.K. Zee, J.E.L. Wong, Singapore's health-care system: key features, challenges, and shifts, Lancet 398 (10305) (Sep. 2021) 1091–1104, https://doi.org/10.1016/S0140-6736(21)00252-X.
[40] M. Britnell, In Search of the Perfect Health System, Bloomsbury Publishing, 2015.
[41] Health Research by Location, Available: https://www.healthdata.org/research-analysis/health-by-location/profiles. (Accessed 06 January 2025).
[42] P.L. Yin, S. Verma, C.S. Ann, Outcomes of the Early Psychosis Intervention Programme (EPIP), The Singapore Family Physician, Singapore, 2013.
[43] T.J. Betjeman, S.E. Soghoian, M.P. Foran, mHealth in sub-Saharan Africa, Int J Telemed Appl 2013 (2013) 1–7, https://doi.org/10.1155/2013/482324.
[44] G.S. Bloomfield, R. Vedanthan, L. Vasudevan, A. Kithei, M. Were, E.J. Velazquez, Mobile health for non-communicable diseases in Sub-Saharan Africa: a systematic review of the literature and strategic framework for research, Glob. Health 10 (1) (2014) 49, https://doi.org/10.1186/1744-8603-10-49.
[45] E.F. Krah, J.G. de Kruijf, Exploring the ambivalent evidence base of mobile health (mHealth): a systematic literature review on the use of mobile phones for the improvement of community health in Africa, Digit Health 2 (Jan) (2016), https://doi.org/10.1177/2055207616679264.
[46] M.J. Annear, J. Otani, J. Sun, Experiences of Japanese aged care: the pursuit of optimal health and cultural engagement, Age Ageing 45 (6) (Nov. 2016) 753–756, https://doi.org/10.1093/ageing/afw144.
[47] Japan Can Teach the World a Better Way to Age, Available: https://www.washingtonpost.com/opinions/2023/08/15/japan-elderly-care-dignity-respect/. (Accessed 06 January 2025).
[48] More Japanese Households with Seniors Caring for Seniors, Available: https://www.nippon.com/en/japan-data/h01733/. (Accessed 06 January 2025).
[49] A. Szczepura, H. Masaki, D. Wild, T. Nomura, M. Collinson, R. Kneafsey, Integrated long-term care 'neighbourhoods' to support older populations: evolving strategies in Japan and England, Int. J. Environ. Res. Publ. Health 20 (14) (Jul. 2023) 6352, https://doi.org/10.3390/ijerph20146352.
[50] M.Y. Chan, et al., Remote postdischarge treatment of patients with acute myocardial infarction by allied

[48] More Japanese Households with Seniors Caring for Seniors, Available: https://www.nippon.com/en/japan-data/h01733/. (Accessed 06 January 2025).
[49] A. Szczepura, H. Masaki, D. Wild, T. Nomura, M. Collinson, R. Kneafsey, Integrated long-term care 'neighbourhoods' to support older populations: evolving strategies in Japan and England, Int. J. Environ. Res. Publ. Health 20 (14) (Jul. 2023) 6352, https://doi.org/10.3390/ijerph20146352.
[50] M.Y. Chan, et al., Remote postdischarge treatment of patients with acute myocardial infarction by allied health care practitioners vs standard care, JAMA Cardiol 6 (7) (Jul. 2021) 830, https://doi.org/10.1001/jamacardio.2020.6721.
[51] R.V. Milani, C.J. Lavie, Health care 2020: reengineering health care delivery to combat chronic disease, Am. J. Med. 128 (4) (Apr. 2015) 337–343, https://doi.org/10.1016/j.amjmed.2014.10.047.
[52] Patients Enrolled In NHGP's Teamlet Care Model Show Improved Diabetic Control, Available: https://www.nhgp.com.sg/Lists/News/Attachments/7/mr07.pdf. (Accessed 06 January 2025).
[53] World Bank and WHO: Half the World Lacks Access to Essential Health Services, 100 Million Still Pushed Into Extreme Poverty Because Of Health Expenses, Available: https://www.who.int/news/item/13-12-2017-

world-bank-and-who-half-the-world-lacks-access-to-essential-health-services-100-million-still-pushed-into-extreme-poverty-because-of-health-expenses. (Accessed 06 January 2025).
[54] Rethinking Health Care from a Global Perspective: American Complexity, Available: https://www.commonwealthfund.org/blog/2024/rethinking-health-care-global-perspective-american-complexity. (Accessed 06 January 2025).
[55] Extent of Self-Reported Unmet Need for Health Care Services in Different Sub-Groups Of Population, Available: https://www.who.int/data/gho/indicator-metadata-registry/imr-details/855. (Accessed 06 January 2025).

Systematic analysis of opportunities for healthcare transformation

2

2.1 Opportunities for healthcare transformation

Transformation of healthcare will mean different things to different people. There are the basic philosophies and strategies that have transformational potential and may change the way we think about healthcare systems. We will first make a list of some of the main contenders as "change methodologies" and later look at tools that can guide us in where the greatest value from transformation might be obtained and where technology and AI have the greatest potential for impact:

1. Shifting of focus and priorities. Such methods include philosophies such as **value-based care,** which concentrates on getting the most value, in the form of patient outcomes and cost-effectiveness, from all investments and processes. This is a fundamental principle and will underlie much of what we discuss in this book.
2. Shifting **from treating illness to prevention** is another technique for which we will lay out some data and implications in later parts of the book.
3. Changing of **public policy** with altered provisions for **payments, entitlements, and benefits.** These may seek to make healthcare more accessible or may reduce underuse and discourage overuse of the system. We have already discussed some of these principles, but many of them are at the will of basic policy decisions by governments, providers, employers, etc., and are influenced by affordability, values, politics, etc.
4. Emphasis on **public health.** Public health is the sum of the entire population's outcomes, so taking a public health perspective is often useful in creating priorities and allocating resources. This is for several reasons. Some people, if not actively sought out, are left outside the system. Programs such as vaccinations, screenings, and education prevent ill-health that extracts a burden from the entire population, thereby impacting societal well-being and overall economic productivity. The COVID-19 pandemic provided good examples of the importance of public health management.

health management.

5. Improving **efficiency of the healthcare system.** There are many sources of inefficiency and waste in healthcare systems, ranging from missed preventative opportunities, inappropriate and inadequate treatments, lack of information in the right place at the right time, and suboptimal resource allocation.
6. Shift to **patient-centricity** which "focuses the attention on patient's beliefs, preferences, and needs, in contrast to physician-centered care," see for example [1]. This includes greater emphasis

Healthcare Transformation using Artificial Intelligence. https://doi.org/10.1016/B978-0-443-28969-9.00004-2

on proactive measures for activity and diet, mental wellness, home- and mobile-based methods of monitoring and management of chronic diseases, helping seniors living alone, etc.

Several healthcare systems have effectively illustrated these opportunities, as we have already seen in a range of innovations from different countries in the last chapter. We will see more examples in specific settings in Part 3, including innovative ways of managing services at Intermountain Health, informative technology exploitations at Ochsner Health, transformations being deployed in Singapore, and some quite different approaches in emerging countries.

We already told the story in the Introduction of Haven Healthcare, Walmart, Optum, etc. and how they found pure plays "at the edge" hard to deliver and sustain as they ventured into the retail and telehealth space. It is worth asking how these very successful businesses could make such radical reversals in the space of a few years. What were they missing in the strategic sustainability of their attempts to standardize (and sometimes commoditize) elements, even though they acted in the general spirit of Innovator's Prescription's Value-Added Providers (VAPs)? The CVS MinuteClinic that we cited as a VAP has been relatively successful, with presence in 1100 stores, but even they have announced some closures recently [2] and have been criticized for not encouraging a long-term announced some closures recently [2] and have been criticized for not encouraging a long-term relationship to develop with patients [3]. The answer is not simple but may center on the importance of taking a longitudinal and long-term view of patient treatment methodologies and resulting outcomes [4]. As we will see later in this chapter, chronic diseases, as large precursors of major disease, need to be the target of sustained and nonepisodic approaches. The same is true of hospitalization which incurs the largest part of overall costs.

2.2 A systems approach to understanding transformation

Automation, technology, and AI can be used to address the above opportunities for transformation. We have already argued previously why information is a powerful lever in solving many healthcare problems. To zero in on them some more, we examine a model in Fig. 2.1 of how members of the population move around and interact with the healthcare system.

People enter the healthcare system by being born, immigrating, or perhaps signing up with a healthcare system for services. But they spend most of their time at home and in their community where their actions and lifestyle control most of their outcomes (see Section 2.4). Eventually everyone falls ill, or needs medical advice, and makes contact with the healthcare system. This could happen at a primary care facility, perhaps a public clinic, via a visit to a school nurse or other health worker, via a telehealth consultation, or via a call to a help center. Usually, the individual returns home and gets better. In some cases, they may progress to specialist care.

And eventually most patients end up receiving some tertiary care, from a specialist or hospital, and this is where things get serious, and costs may skyrocket. They may have an outpatient experience, be admitted into a general or specialist ward, or even an ICU. They may undergo procedures or surgeries. But eventually most are discharged to go home, perhaps with some recommended follow-up with primary care or a specialist. Eventually people may leave the system to join a new one or may pass away.

The patient flow model in Fig. 2.1 is a fundamental and standard way of analyzing any service system, see for example the queueing networks section of [5]. And there are only a few ways that we can improve or transform any such system. When applying such an analysis to a healthcare system, "improve" will mean improving the journey or experience as people move through the system. For

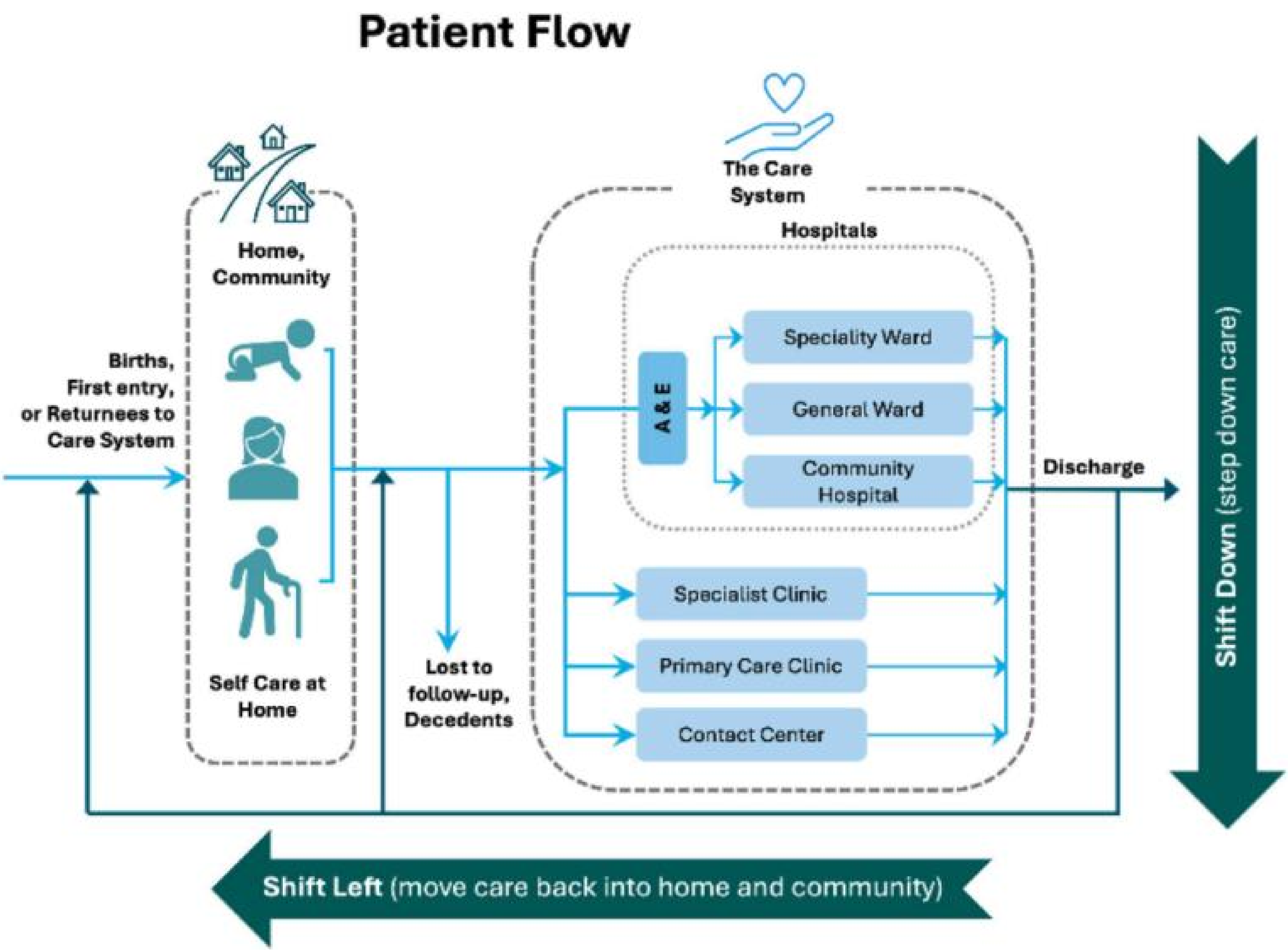


FIGURE 2.1

Patient flow between home, community, and the care system.

patients, they will experience waiting times, costs, outcomes, etc. From the perspective of the system, we mean performance in the form of quality, cost, and public health outcomes.

Methods of improvement include:

Methods of improvement include:

1. Changing the service that is received in each of nodes (blocks) within the system. This could involve changing the actual treatment or improving the quality or productivity of an existing treatment, as discussed in the prior chapter. It involves the cost incurred (time spent and resources used) and services received in each node of the system. Since resources may be constrained, each of the nodes may also involve *waiting times*, which could be spent in the facility or waiting at home for a future appointment. Nobody likes waiting.
2. Changing the location (node) or nature of services at which people receive a service. That is, they may be routed to a different physical location or setting (e.g., moving a specific service from hospital to a clinic or to home) within the system.
3. Changing the flows within the system, which could involve avoiding visiting a facility because the service could be received elsewhere (e.g., self-managed or at home), or because the visit is avoided altogether.

As we go through transformations, we will see many instances and versions of these improvements being applied. But there are fundamental principles at play as we work to transform and optimize this system. One we will call *shift-left* where, referring to the previous diagram, we move treatments to the left, meaning toward the home or community and conduct them through behavior change, self-care, remote care, or perhaps telecare; we will give examples shortly. One of the most currently discussed and important examples of shift-left is hospital-at-home, a major and impactful transformation being carried out throughout the world. It allows for outright avoidance of hospital admissions, in some cases. And it can shorten length-of-stay due to early discharge from hospital to home. It may include provision of home care, which is the equal of hospital care for many conditions, in the comfort of home, using a combination of technology for monitoring/interventions and home visits by professionals.

professionals.

Another transformation we call *shift-down*, as illustrated in Fig. 2.1, involves lowering the acuity of the care settings. This could mean moving follow-ups from a specialist to a primary care, moving from physician to an allied health professional (as discussed in the last chapter), or perhaps use of a visiting nurse. Or it could mean moving patients from a specialized or acute hospital experience to a *general hospital* experience. These models have the advantage that a more holistic care experience can be achieved, and in some cases, costs lowered. It is well recognized that many older patients have multiple conditions (sometimes called comorbidities), and having a *one-care-team* in a clinic or hospital can provide more holistic care, including avoiding problems such as inappropriate polypharmacy (excessive drugs or drug interactions), etc.

Several examples in Part 3 of this book will show examples of shift-left and shift-down.

2.3 Using data, information, and real facts to identify and focus targets of change in healthcare

Before we get to AI in Part 2, we need to further establish some additional facts and data about healthcare, as it will allow us to set the stage for understanding the potential of AI.

Healthcare is almost always a large and complex system being applied to another large and complex system - the society - and to try to do something to improve it you have to be strategic. And that means picking our battles and focusing on a few of the biggest levers that can make the most change: changes must be both executable and cost-effective. Like any other business investment, we must not just shoot for the return but consider the expected value of the return divided by the cost and probability of success in making that change, that is, the return on investment (ROI).

Fortunately, as previously described, healthcare has masses of good data (especially claims and financial data) that can give us some hints as to where to focus. Medicine also has a strong tradition of measurement, data collection, and evidence-based science. And since a lot of capable and dedicated people have worked for a long time to build and improve healthcare systems, it is worth asking them what they see as the biggest shortcomings and opportunities for improvement. The extent of both needs and potential for improvement can be captured, to first order, by focusing on data captured on lifespan, healthspan (the length of time we are healthy), population change, the incidence of disease, suffering and disability, and early deaths.

I will use an example and data from the Singapore Healthcare system because that is where I am working. Data and insights into the country's systems can be found in three excellent books [6–8]. In

Singapore, life expectancy is now 82.3 versus 80.7 years in the OECD. And the Singapore system delivers excellent care at a cost which, as a fraction of GDP, is smaller than other high-income countries. Why is this? A great example of human incentives: in Singapore, the patient is asked to pay something for almost every treatment and medication they receive. The use of out-of-pocket fees is supposed to be an effective way to limit consumption and prevent *over-use* of healthcare resources and should have that effect.

But it has also been shown that a negative relationship exists between out-of-pocket expenses and health outcomes. For evidence in lower income countries, see Ref. [9]. In higher income countries, out-of-pocket expenses for the employed and middle-income groups have been rapidly increasing. Higher out-of-pocket expenses have been linked to poorer adherence and delays in care, see the overview in Ref. [10]. We certainly should always be on the lookout for *under-use* as we saw in Chapter 1. An example of underuse is when patients have a chronic disease (such as diabetes, hypertension, dyslipidemia), and they either do not know about it, or they ignore or postpone control measures. These patients sometimes first appear in the healthcare system when they show up in the emergency room with a heart attack, stroke, or other illness. Most of the organ damage may already be done, and it now becomes difficult and expensive to treat.

With life expectancies rising worldwide, birth rates decreasing, and immigration moderating, the net effect is that the population distribution in most countries is quickly becoming more aged. And along with that trend comes the "diseases of age." In Singapore, we expect that by 2030 one in seven patients will have diabetes (which plays havoc with the cardiovascular system), and we are expecting a 60%–110% increases in the incidence of heart attacks, cancer, and stroke.

It is worth digging in, and there are two very powerful measures promulgated by WHO that are very useful as measures of world health trends. The first of these is *Years of Life Lost*, or YLLs, which sum up all the premature deaths and then attributes them proportionally to primary diseases or accidental causes. These are all collected in a wonderful website at the WHO Global Health Observatory [11]. The World Bank classifies countries into low-, lower-middle-, upper-middle-, or high-income [12]. For all but some low-income countries, the predominant attribution of YLLs tends to be cardiovascular diseases. For example, in Singapore, this figure is 28.5%, compared to a worldwide figure of 20%. Back to strategy: what actions could we take *that would have high ROI*? Cardiovascular diseases are relatively modifiable by behavior change, and there are widely available treatments such as low-cost medications. Cancer is also quite a high cause of YLLs, but it is not so modifiable, and the prevention and treatment of cancer, while making progress, has been much harder to achieve [13].

Singapore, life expectancy is now 82.3 versus 80.7 years in the OECD. And the Singapore system delivers excellent care at a cost which, as a fraction of GDP, is smaller than other high-income countries. Why is this? A great example of human incentives: in Singapore, the patient is asked to pay something for almost every treatment and medication they receive. The use of out-of-pocket fees is supposed to be an effective way to limit consumption and prevent *over-use* of healthcare resources and should have that effect.

But it has also been shown that a negative relationship exists between out-of-pocket expenses and health outcomes. For evidence in lower income countries, see Ref. [9]. In higher income countries, out-of-pocket expenses for the employed and middle-income groups have been rapidly increasing. Higher out-of-pocket expenses have been linked to poorer adherence and delays in care, see the overview in Ref. [10]. We certainly should always be on the lookout for *under-use* as we saw in Chapter 1. An example of underuse is when patients have a chronic disease (such as diabetes, hypertension, dyslipidemia), and they either do not know about it, or they ignore or postpone control measures. These patients sometimes first appear in the healthcare system when they show up in the emergency room with a heart attack, stroke, or other illness. Most of the organ damage may already be done, and it now becomes difficult and expensive to treat.

With life expectancies rising worldwide, birth rates decreasing, and immigration moderating, the net effect is that the population distribution in most countries is quickly becoming more aged. And along with that trend comes the "diseases of age." In Singapore, we expect that by 2030 one in seven patients will have diabetes (which plays havoc with the cardiovascular system), and we are expecting a 60%–110% increases in the incidence of heart attacks, cancer, and stroke.

It is worth digging in, and there are two very powerful measures promulgated by WHO that are very useful as measures of world health trends. The first of these is *Years of Life Lost*, or YLLs, which sum up all the premature deaths and then attributes them proportionally to primary diseases or accidental causes. These are all collected in a wonderful website at the WHO Global Health Observatory [11]. The World Bank classifies countries into low-, lower-middle-, upper-middle-, or high-income [12]. For all but some low-income countries, the predominant attribution of YLLs tends to be cardiovascular diseases. For example, in Singapore, this figure is 28.5%, compared to a worldwide figure of 20%. Back to strategy: what actions could we take *that would have high ROI*? Cardiovascular diseases are relatively modifiable by behavior change, and there are widely available treatments such as low-cost medications. Cancer is also quite a high cause of YLLs, but it is not so modifiable, and the prevention and treatment of cancer, while making progress, has been much harder to achieve [13].

and treatment of cancer, while making progress, has been much harder to achieve [13].

The second measure of great interest relates to health span. Many people end up spending their last 10+ years or more of life disabled and not living a life with high quality, enjoyment, or dignity. This is measured by the second WHO metric, namely *years lived with disabilities*, or YLDs. When considering YLDs, we find another relatively neglected disease, mental health (e.g., anxiety and depression), that can be treated but remains relatively underserved. In many countries, and especially in Asia, mental health suffers heavily from stigma. Mental health is often demoted to second place relative to physical health, as can be seen by the low number of mental health professionals in many countries. And it has devastating consequences which include the increasing rates of suicide in youth, perhaps exacerbated by many of life's increasing social pressures. But suicides are also increasing among the aged, perhaps related to pain, disability, and loneliness. In Singapore, mental health, at 20%, is one of the highest causes of YLDs. Close to this are musculoskeletal disorders, with high prevalence in older residents, which are harder to treat.

2.4 The predominant determinants of health outcomes

Let us dig even deeper on the causes. The landmark work of McGinnis [14] and Schroeder [15] is sometimes known as the **40/30/20/10 rule** for loss of YLLs or causes of premature death. It is based on meta-analysis of thousands of research papers. In the case of these two references, the data are focused on the US, but it would be expected to have similar form in other countries (however, see Chapter 13 for emerging countries). When quoted, it often varies somewhat in the quoted percentages but tends to have this form:

1. **40%** of YLLs are attributable to **lifestyle**, that is, diet, activity, smoking, stress, etc. These causes can in principle be changed. Although changing such human behavior is often difficult, awareness and education continue to play a key role as many people do not even know or realize that they

can in principle be changed. Although changing such human behavior is often difficult, awareness and education continue to play a key role as many people do not even know or realize that they have to change.

2. **30%** is attributable to **genetics**, or "how we chose our parents." This is much harder to change, but when you know you are more exposed to genetic risks such as familial high cholesterol or breast cancer you can take extra steps to guard against and manage your disease.
3. **20%** is attributable to **social and environmental factors.** These include the well-known effects of a poor environment, including poor air quality and presence of water-borne diseases. These factors contribute to a stubbornly high correlation between low income and poor health outcomes. Sometimes, this is attributed to having poor access to healthcare, but even in countries where quality healthcare is freely available, the effect of social factors persists. For example, in the previous chapter, we cited decreased lifespans in lower SES groups. Some believe that this might be a consequence of education, or perhaps it is related to the stresses of feeling insecure or "kicked down" by the society.
4. And finally, **10%** is due to the **healthcare** we receive. What? Only 10%? Yes, the fact is, as explained above, by the time we receive treatment for the complications of a disease, that disease has often progressed to a point where there are limited treatment options. The 40% and 20% items above are far more changeable and addressable by our systems. However, we should not underestimate the importance of this critical 10%: once we fall sick it will be a key determinative factor for our outcome.

The 40/30/20/10 rule teaches us a lot about what we can do to transform healthcare. It is about "a day in the life of a patient" which we show graphically in Fig. 2.2.

Previously we described the strategies of shift-left to move people back into the community, and in later chapters, we will describe innovative tech projects that focus on shift-left in management of chronic diseases such as diabetes and hypertension. If these chronic diseases have already progressed, we can further ask whether we could move and improve their care in the primary care setting and at home.

In fact, in most heath care systems, 70%–80% of costs are incurred in the tertiary or hospital setting. And not only that, frequently 60%–70% of all medical costs are incurred in the last 6 months of life. Thus, shift-down is an equally or more powerful method of care where we push care from the acute hospital into lower acuity hospital settings, and both strategies apply when we push from specialist clinics back into lower-acuity outpatient settings including primary care.

And if tech can help, we should use it. In Part 3, we will also talk about how we can manage both chronic diseases and mental health, especially through self-help at home and in the community. And

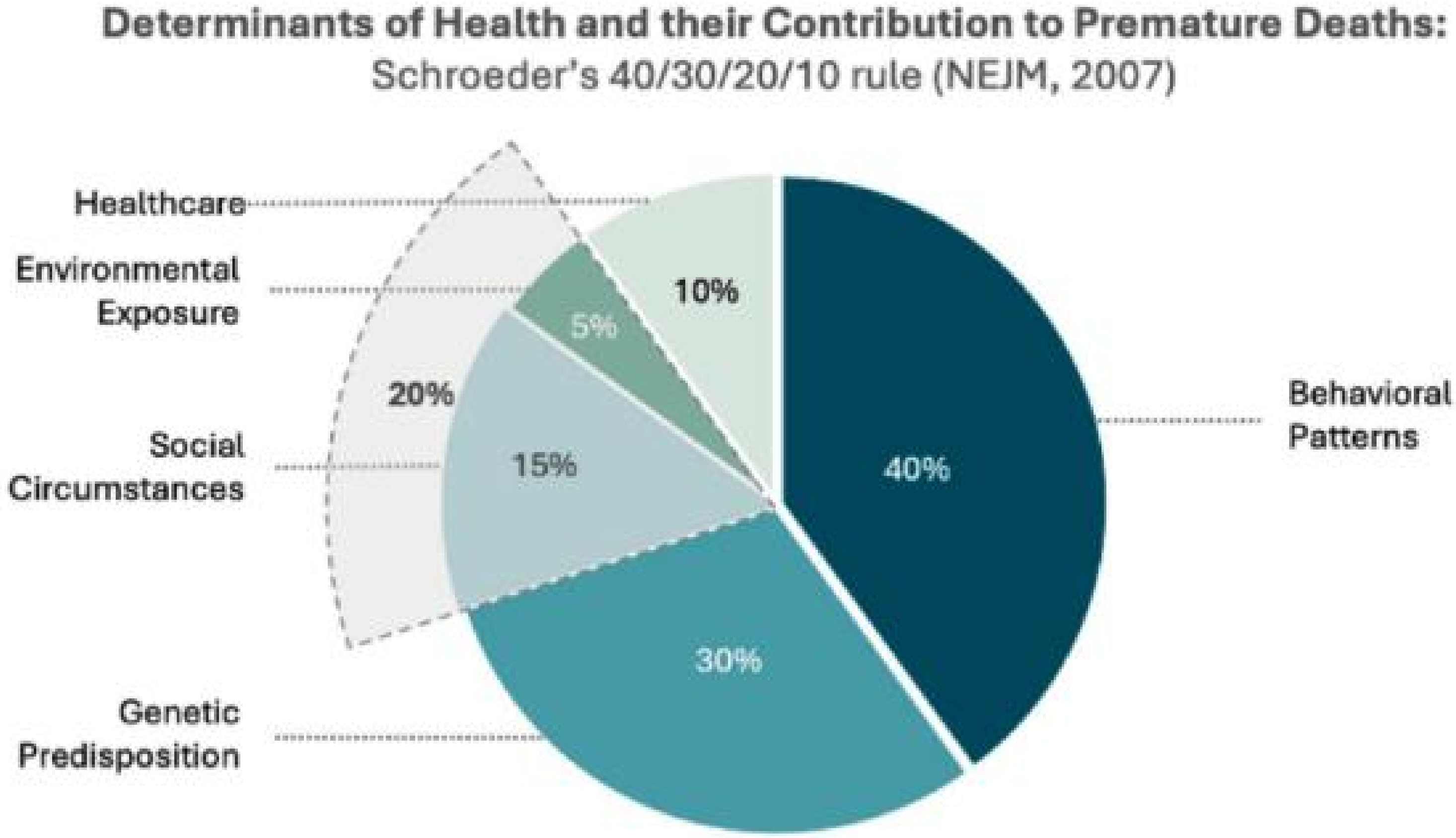


FIGURE 2.2

Determinants of health and their contributions to premature deaths.

where diseases are more serious, again we try to move from hospital and medicalized settings back into the community with peer support, and use trained peers, counselors, therapists, and social workers who can often also deal with the root causes of unhealthy behaviors and mental stresses.

2.5 Putting it all together

As we have seen in Chapter 1, there are considerable differences in how different healthcare systems perform and in how well different business models (fee-for-service, fee-for outcome, and fee-for-membership) work. In this chapter, we have begun to see how systematic approaches will help us know where to focus within a healthcare system and the population that it serves. We are guided by the root causes of poor outcomes, which are often due to lifestyle factors or activities that are *not yet* a sufficient focus of the healthcare system. In the next chapter, we will see how data, information, and technology may be able to help. Then, after introducing AI in Part 2, we will then embark on a journey in Part 3 around different venues in a patient's daily life, settings where they can get help, and look at ongoing and successful approaches to transforming these processes with AI.

References

[1] R. Jayadevappa, Patient-centered outcomes research and patient-centered care for older adults, Gerontol Geriatr Med 3 (Jan. 2017), https://doi.org/10.1177/2333721417700759.

[2] S. Masunaga, These are the CVS MinuteClinics that Will Close Around Los Angeles, Available: https://www.latimes.com/business/story/2024-01-31/cvs-to-close-25-minuteclinics-los-angeles-area. (Accessed 11 January 2025).

[3] MinuteClinic—Wikipedia, Available: https://en.wikipedia.org/wiki/MinuteClinic. (Accessed 6 January 2025).

[4] The End of Walmart Health, and Retail Health's Demise?, Available: https://hospitalogy.com/articles/2024-05-02/end-of-walmart-health-retail-health-implications/. (Accessed 6 January 2025).

[5] Queueing theory—Wikipedia, Available: https://en.wikipedia.org/wiki/Queueing_theory. (Accessed 6 January 2025).

[6] W.A. Haseltine, Affordable Excellence: The Singapore Healthcare Story, Brookings Institution Press, 2013. Available: https://rowman.com/ISBN/9780815725268/Affordable-Excellence-The-Singapore-Healthcare-Story>. (Accessed 13 January 2025).

[7] M. Britnell, In Search of the Perfect Health System, Bloomsbury Publishing, 2015.

[8] J. Lim, Myth or Magic: the Singapore healthcare System, vol 2013, Select Publishing Singapore, 2013.

[9] V.M. Qin, et al., The impact of user charges on health outcomes in low-income and middle-income countries: a systematic review, BMJ Glob. Health 3 (Suppl. 3) (Jan. 2019) e001087, https://doi.org/10.1136/bmjgh-2018-001087.

[10] B. Z. S.A. Glied, Catastrophic Out-of-Pocket Health Care Costs: A Problem Mainly for Middle-Income Americans with Employer Coverage, Available: https://www.commonwealthfund.org/publications/issue-briefs/2020/apr/catastrophic-out-of-pocket-costs-problem-middle-income. (Accessed 6 January 2025).

[11] The Global Health Observatory, Available: https://www.who.int/data/gho. (Accessed 6 January 2025).

[12] World Bank Country and Lending Groups, Available: https://datahelpdesk.worldbank.org/knowledgebase/articles/906519. (Accessed 6 January 2025).

[13] S. Mukherjee, The Emperor of All Maladies: A Biography of Cancer, Simon and Shuster, 2010.

[14] J.M. McGinnis, Actual causes of death in the United States, JAMA, J. Am. Med. Assoc. 270 (18) (Nov. 1993) 2207, https://doi.org/10.1001/jama.1993.03510180077038.

[3] MinuteClinic—Wikipedia, Available: https://en.wikipedia.org/wiki/MinuteClinic. (Accessed 6 January 2025).
[4] The End of Walmart Health, and Retail Health's Demise?, Available: https://hospitalogy.com/articles/2024-05-02/end-of-walmart-health-retail-health-implications/. (Accessed 6 January 2025).
[5] Queueing theory—Wikipedia, Available: https://en.wikipedia.org/wiki/Queueing_theory. (Accessed 6 January 2025).
[6] W.A. Haseltine, Affordable Excellence: The Singapore Healthcare Story, Brookings Institution Press, 2013. Available: https://rowman.com/ISBN/9780815725268/Affordable-Excellence-The-Singapore-Healthcare-Story>. (Accessed 13 January 2025).
[7] M. Britnell, In Search of the Perfect Health System, Bloomsbury Publishing, 2015.
[8] J. Lim, Myth or Magic: the Singapore healthcare System, vol 2013, Select Publishing Singapore, 2013.
[9] V.M. Qin, et al., The impact of user charges on health outcomes in low-income and middle-income countries: a systematic review, BMJ Glob. Health 3 (Suppl. 3) (Jan. 2019) e001087, https://doi.org/10.1136/bmjgh-2018-001087.
[10] B. Z. S.A. Glied, Catastrophic Out-of-Pocket Health Care Costs: A Problem Mainly for Middle-Income Americans with Employer Coverage, Available: https://www.commonwealthfund.org/publications/issue-briefs/2020/apr/catastrophic-out-of-pocket-costs-problem-middle-income. (Accessed 6 January 2025).
[11] The Global Health Observatory, Available: https://www.who.int/data/gho. (Accessed 6 January 2025).
[12] World Bank Country and Lending Groups, Available: https://datahelpdesk.worldbank.org/knowledgebase/articles/906519. (Accessed 6 January 2025).
[13] S. Mukherjee, The Emperor of All Maladies: A Biography of Cancer, Simon and Shuster, 2010.
[14] J.M. McGinnis, Actual causes of death in the United States, JAMA, J. Am. Med. Assoc. 270 (18) (Nov. 1993) 2207, https://doi.org/10.1001/jama.1993.03510180077038.
[15] S.A. Schroeder, We can do better—improving the health of the American people, N. Engl. J. Med. 357 (12) (Sep. 2007) 1221–1228, https://doi.org/10.1056/NEJMsa073350.

The role of information and technology in transforming industries and how it could transform healthcare

3

3.1 Lessons from other industries as they have used information to transform: Methods of industry transformation

As we described in the Introduction and Chapter 1, creation of successful healthcare business models is not easy. There have been many miscalculations and failed strategies. The same has happened in other industries. Just adding technologies and artificial intelligence (AI) into the mix will not necessarily provide success.

Before moving on to the role of information, data, and technology in healthcare, we pause to gather some lessons from technology-aided transformations in other industries that may inform us in our quest.

Perhaps, the two biggest global transformations of the last 100 years have been the *information revolution* and the *globalization of industry and trade*.

The information revolution has revolved around the development of computing and telecommunications, which has brought the world's information to our fingertips and made it constantly available through small devices that we always keep with us. The information revolution has also aided the transformation of large parts of the workforce into knowledge-based services, of which healthcare is now one of the largest sectors.

The globalization of industry and trade has been made possible by improvements in the productivity of agriculture that has released massive numbers of workers from farming into manufacturing. Barely 100 years ago, over half of the world's population was consumed in the production of food, but now huge numbers of people have moved into manufacturing and knowledge work, of which the most notable example has been the movement of hundreds of millions of Chinese citizens into these jobs and into the middle class.

Therefore, we choose just two industries from which to extract lessons that may inform us in our quest for transformation in healthcare: information technology and manufacturing.

The **information revolution** has yielded massive progress in industries such as media, banking and finance, retail, etc. While some impact has been felt in healthcare (e.g., the digitalization of medical records), the information revolution has only just started to improve human health.

In Annex A, we give more details about the transformation that has occurred in the information

In Annex A, we give more details about the transformation that has occurred in the information technology industry. We normally think of the relentless pace of microelectronics technology as the driver of impact. But a key observation is that progress of the base technology has enabled, *but not*

Healthcare Transformation using Artificial Intelligence. https://doi.org/10.1016/B978-0-443-28969-9.00005-4

always delivered, industry transformations.[1] To fully deliver on its potential, technology has to be enveloped in new ways of making its potential accessible and easily usable. Specifically, true disruptions have usually needed to be accompanied by innovations in the user interface to information, how people access information—it then has the chance to change *business processes* and *business models*.[2] The same will be true of healthcare.

For example, while semiconductor, storage, and communications technology pushed forward, it was the transition from mainframe to departmental computing, and then to personal computing, and even further to computing in our pockets that changed the way we work and live. Similarly, the Internet had existed for 20+ years before the browser. It was only when the browser made the Internet easy enough to use by everyone (aided by personal computing) that its impact on retail, banking, media, etc. occurred. Even though AI has been developing for many years, it only received widespread adoption at the end of 2022 when it became easily usable through a free text natural language bot interface. Similarly, it may happen in healthcare that only when the information is easily brought to the point of care (through improved care interfaces and processes) and conveniently enters our daily lives (e.g., through wearables, apps, or ubiquitous computing) that healthcare will enable the revolution that we aspired to in Chapter 1.

Another industry effect that we describe in Annex A concerns *business models*. As an industry has become more complex, new business models may emerge, for example subscription and service models. While the details of the models may differ, it could be argued that models of capitation (and what we earlier described as fee-for-membership) are novel and analogous business models that change access costs and behaviors to provide better and more sustainable outcomes.

The second global transformation we consider results in part from the decreased portion of human labor dedicated to agriculture (over a period of 100 years), down from about 50% to now 1%–2% in developed countries, although somewhat lagging in less-developed countries. The rise of **manufacturing** has replaced those jobs released from agriculture and changed the world through

manufacturing has replaced those jobs released from agriculture and changed the world through globalization and trade. And so, in Annex B, we consider the key lessons of manufacturing. Manufacturing is all about how complex entities flow through, and are served, by complex systems, involving both humans and machines. This is analogous to, and holds lessons for, healthcare, particularly around transformation of what must inherently remain human-centered business processes.

We already referenced in Chapter 1 how global manufacturing might offer useful lessons for healthcare systems. One of the key observations from manufacturing is that **superior business processes**, where humans play a key part and there are good human factors (i.e., interactions between humans and humans and humans and machines), are essential to deliver quality or value from this complex business process. This turns out to be more important than the brute force of an automation, or its impact just via throughput or productivity. Business process change was a key factor in the triumph

[1]"In 1987 Robert Solow, a Nobel laureate, famously quipped that "you can see the computer age everywhere but in the productivity statistics." Only in the late 1990s did the economic transformation at last materialize, leading Solow to acknowledge—three decades after the initial exuberance—that computers had begun to reshape the economy." From an article asking when AI might deliver similar benefits: https://www.economist.com/finance-and-economics/2024/11/21/computers-unleashed-economic-growth-will-artificial-intelligence.

[2]Business processes and business models were defined in footnote 5 of Chapter 1.

of Japanese manufacturing culture versus a "technology push" of American manufacturing culture in the 80 and 90s. A key lesson for healthcare is that needed information must be at the right place at the right time to impact and aid its consumption by a clinician delivering care or by a human who may have to change their lifestyle to improve their health.

For example, if we come back to the patient flow schematic of Chapter 2's Figure 2.1, the systems approach diagram intuitively suggests that if you can reroute a patient to a lower acuity setting (which could be the home) or avoid a visit to the system altogether, it might have greater impact on service quality and costs than if you purely concentrate on improvement of the operation within a service node. That is the strength of business process transformation thinking: recall the example from Chapter 1

That is the strength of business process transformation thinking: recall the example from Chapter 1 where no medical intervention delivered might be the next-best-action.

Thus, a principle can be stated as follows: *A superior business process may beat a local optimization of a unit operation. A superior business process may beat use of technology, applied tactically or locally within a system.*

A second key observation culled from the information revolution concerns **superior business models**. The web created completely new business models that were not enabled and not even contemplated or invented, until way after the technology arrived. For example, search was cool but did not enable massive impact until it revolutionized advertising—hence providing for the ascent of companies such as Google and Meta. Broadband communications bandwidth was more than adequate for some time, but it took the innovation of the streaming business model to overtake media and displace linear TV and the cinema (and to trigger the further buildout of bandwidth). At the time of writing, several currently available tech breakthroughs, AI included, still have not yet developed generally viable or accepted business models. Some of these breakthroughs might well be waiting to be found in healthcare. Businesses where information speed, volume, and complexity outrun human abilities typically end up having their costs dominated by labor. Labor is presently and by far the dominant cost in both IT and healthcare. Therefore, the IT models of outsourcing, automation, and rent-vs-buy, which strike at the heart of labor availability and scaling limitations, also take hold. This is the story of *cloud computing*. The most recent innovation within the information industry, AI, may offer an even larger opportunity for technology-aided transformation, if applied correctly.

Thus, there is something that can even beat the business process principle stated above, and that is a change in *business model*.[2] Both the business model and the business processes will define the way that business is conducted: what an enterprise and its people (the labor force) do day-to-day to accomplish their mission. And how, and on what bases, businesses are contracted or paid to provide a business proposition or service. As we see in the Annexes, the interactions between people and people and people and technology, whether transactions are for products or services and whether there are fixed purchase fees or a for-hire model, change the whole success proposition. If successful, the business model produces transformations that win in the hard knocks of the marketplace. As we have brought out in Chapter 1, particularly the observations around fee-for-service versus fee-for-outcome versus fee-for-membership, business models can be a profound transformative force in healthcare.

In the case of healthcare, superior business models focus largely on competitive positioning, ease-of-access, human incentives, and trial-by-wallet in the marketplace. You could say that customers or patients “vote with their feet.” But it can be even more profound, as in systems like Kaiser Permanente:

by focusing on prevention and long-term maintenance of care, under a single fixed payment regardless of consumption (a form of Capitation), Kaiser manages to create better long-term relationships between the patient and the care system and thereby discourage both underuse and overservicing,[3] as well as increase the emphasis on prevention. A powerful collateral business benefit for Kaiser is their very high client retention.

Value-based care can also be seen as a progenitor of this principle. Christensen, whose Innovator's Prescription we discussed in Chapter 2, liked to tell the story that when he was first taking his ideas around about disruptive technologies, he had a significant wakeup when he discussed them with Andy Grove, cofounder of Intel. Andy told him that it *was not that the technologies were disruptive*, but that they enabled disruptive *business models* that made the new products easier to adopt.

To summarize:

Transformation principles:

1. *A superior business process may beat a local optimization of a unit operation. A superior business process may beat a use of technology, applied tactically or locally within a system.*
2. *A superior business model, including incentives, may beat a change in business process if it shifts behavior of participants toward better use of resources and more effective outcomes.*

And these are lessons that directly affect us in healthcare. We should note that what we have been calling business models are often called "care models" in healthcare. Care models are essentially business models that apply all resources whether they be capital, resources, and information to carry out the business of healthcare, that is, the providing of services to improve health outcomes.

The transformation stories of both the computer and manufacturing industries also harbor another set of important lessons. They show that **human and interface effects often matter more than the intended transactional effects**. To repeat, this comes about because an underlying technology is hidden and not fully utilized until it becomes accessible and easy-to-use. Both industries tell a common story: **human interactions with complex systems, properly aided, far outrun pure automation or technical capacity—the human factors dominate.** The same is true at the patient–clinician interface.

Human effects to be kept in mind in deploying healthcare transformations include:

1. Whether all the information is readily available to make decisions
2. Whether the information can be digested, combined and presented in an efficient way that allows for an optimal decision
3. Whether the treatment decision can be communicated to the patient in an effective and understandable way and in a way that induces trust
4. Whether the treatments will be complied with. For example, medication adherence often runs in the range of 50%–70% for common chronic conditions [1], and patients may not have sufficient understanding, visibility, or awareness of their condition to make the needed changes in their

understanding, visibility, or awareness of their condition to make the needed changes in their daily living.

The above transformation principles are quite general and apply in many industries and in many settings within healthcare, but we end this section with an irreverent twist and a question. In the first principle, we may appear to have downplayed the role of technology in healthcare transformation, that

[3] It is reported that Kaiser has the largest market share in California, at 43% or more, https://www.benefitscafe.com/insurance-companies/.

is largely because systemic effects are usually more important than blind automation, or "automating the cow path" as it has been called. However, every rule-of-thumb has exceptions.

Once in a generation a technology comes along that "changes everything." It creates such a disruption and opportunity for new thinking that it overturns industries and throws all the playing cards up in the air. The Internet of the 1990s was one such case. The Internet upended the retail industry, enabled, and then triggered e-commerce. It has also completely disrupted the advertising industry, where advertisers had little control or insight into where their ads landed. The emergence of digital marketing has created some of the richest and more powerful industries in the world. Internet connectivity has also disrupted media; we are witnessing the end of linear TV and broadcasting, as everything moves to streaming. Rather than being a "tech push" with minor changes to, or automation of, a business process or business model, it has been totally disruptive and sui generis triggers the *creation* of *fundamentally new* business processes and business models.

We now ask whether AI is this generation's similar disruptor. So far technology has mainly allowed for some optimizations and shift-lefts and shift-downs in the healthcare model. But if we look at the discontinuous change of the last few years in AI's capability (particularly advances in machine learning and generative AI), it seems that we are now on the cusp of something different that might just change everything.

The totality of these learnings from other industries will be revisited as we journey around the settings in Part 3 and where technology may be able to significantly impact and even transform healthcare.

nectivity has also disrupted media, we are witnessing the end of linear TV and broadcasting, as everything moves to streaming. Rather than being a "tech push" with minor changes to, or automation of, a business process or business model, it has been totally disruptive and sui generis triggers the *creation* of *fundamentally new* business processes and business models.

We now ask whether AI is this generation's similar disruptor. So far technology has mainly allowed for some optimizations and shift-lefts and shift-downs in the healthcare model. But if we look at the discontinuous change of the last few years in AI's capability (particularly advances in machine learning and generative AI), it seems that we are now on the cusp of something different that might just change everything.

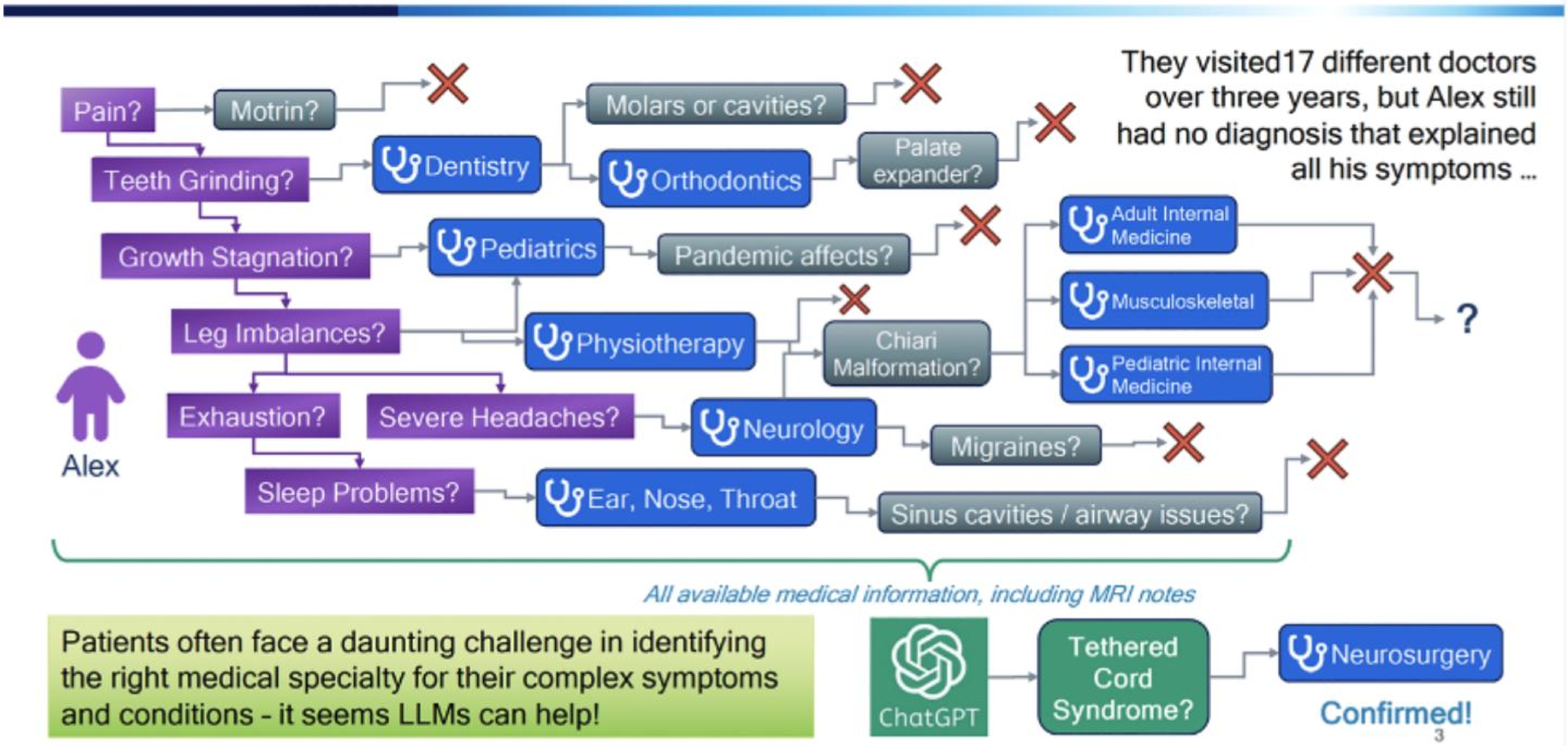
The totality of these learnings from other industries will be revisited as we journey around the settings in Part 3 and where technology may be able to significantly impact and even transform healthcare.

3.2 Technology and healthcare transformation—Why now?

One may ask "why now" for this transformation using information in healthcare? The first is that *we finally have the data*. This comes in several forms, for example, masses of data about longitudinal records and how disease develops and progresses. The implications of this are vast as we will see in later chapters. We have new lab and imaging modalities, also discussed later. Precision medicine is emerging as an opportunity due to an emerging flood of "omics" data, allowing us to better understand the fine structure of the patient. Some data (e.g., wearable devices collecting activity, heart rate, blood oxygen and pressure, blood sugar, etc.) can even be collected continuously. Mobile devices and networks and the *Internet of Things* allow us to immediately see, store, and communicate these data. Electronic records are increasingly used, now by about 90% of physicians in the US [2]. This is made possible by the surge in technologies shown in Fig. 3.1.

The second reason is that recent advances in big data and AI now allow much better processing, understanding, and acting on that data. For a particular patient, we can now access, summarize, and interpret the patient's health records, lab, and image data; we can scan large volumes of the medical literature as an evidence base, predict what will happen to the patient with and without a treatment, and come up with a more informed treatment plan. We can survey entire populations to create epidemiological models of disease and build predictive models through machine learning, not just for the patient but for the population. This approach, called *population health* allows us to look at our entire population and thereby derive polices that most optimally allocate spend so that it optimizes the bottom-line figures of the cost of healthcare, lifespans, health spans, etc.

These methods are rapidly improving both the constitution and usability of the evidence base, as well as supporting better care for an individual patient. The new era of *precision medicine* uses an even broader



A Diagnostic Odyssey: Case involving Tethered Cord syndrome [11]. Figure by Hua Xu, as presented at AI in Health Conference, Rice University, Sep 10–11, 2024 [12].

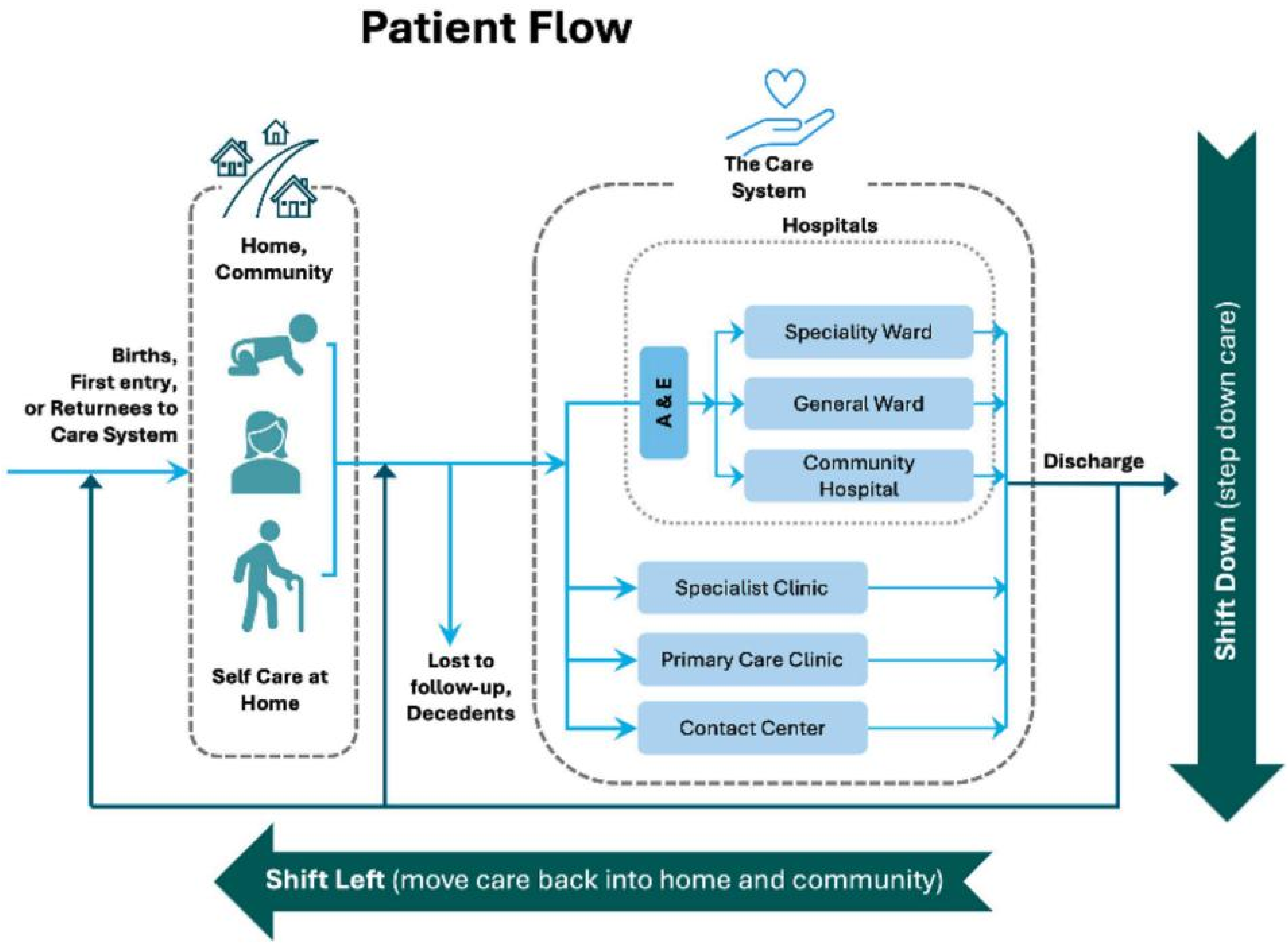


FIGURE 2.1

Patient flow between home, community, and the care system.

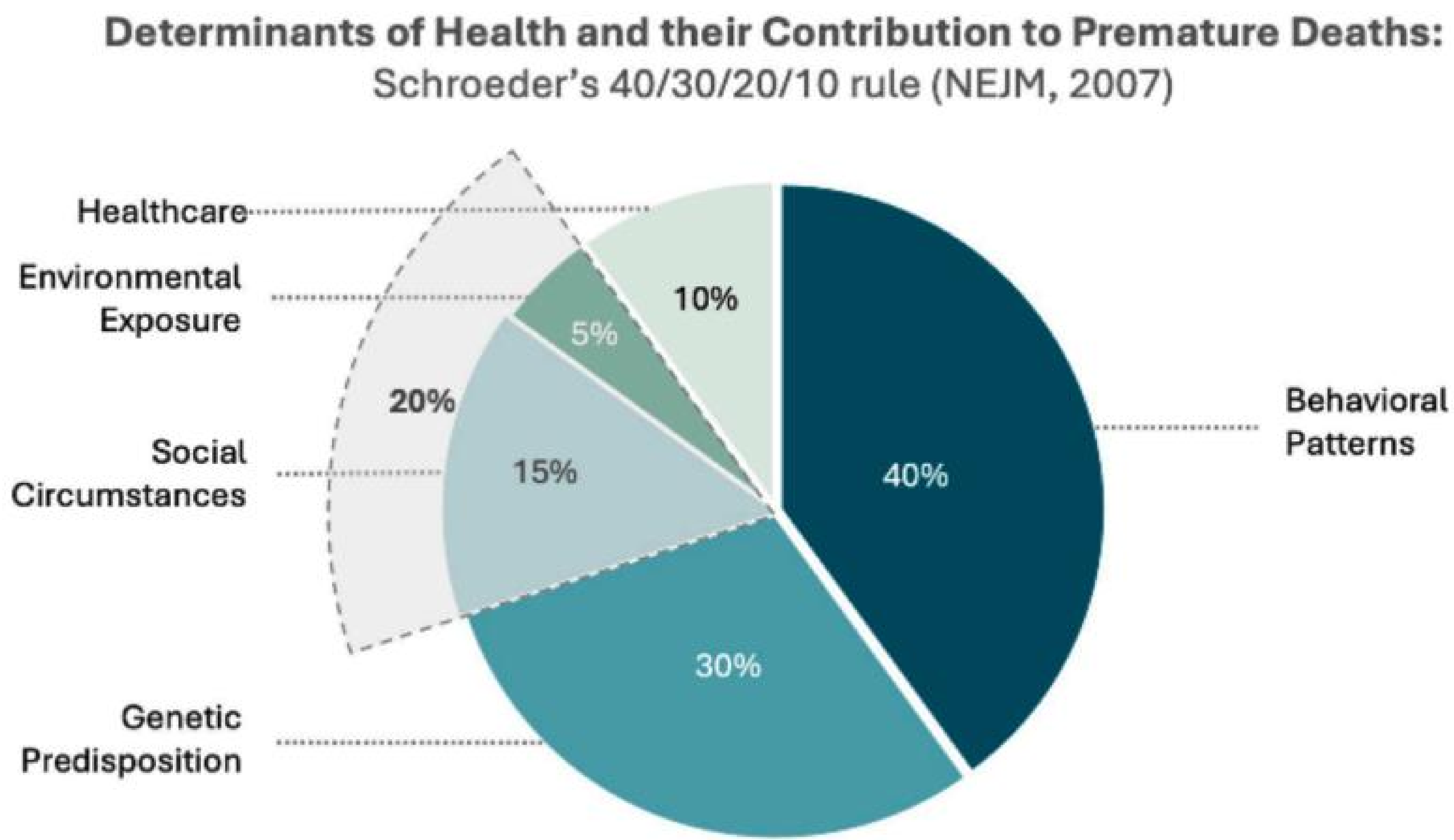


FIGURE 2.2

Determinants of health and their contributions to premature deaths.

where diseases are more serious, again we try to move from hospital and medicalized settings back into the community with peer support, and use trained peers, counselors, therapists, and social workers who can often also deal with the root causes of unhealthy behaviors and mental stresses.